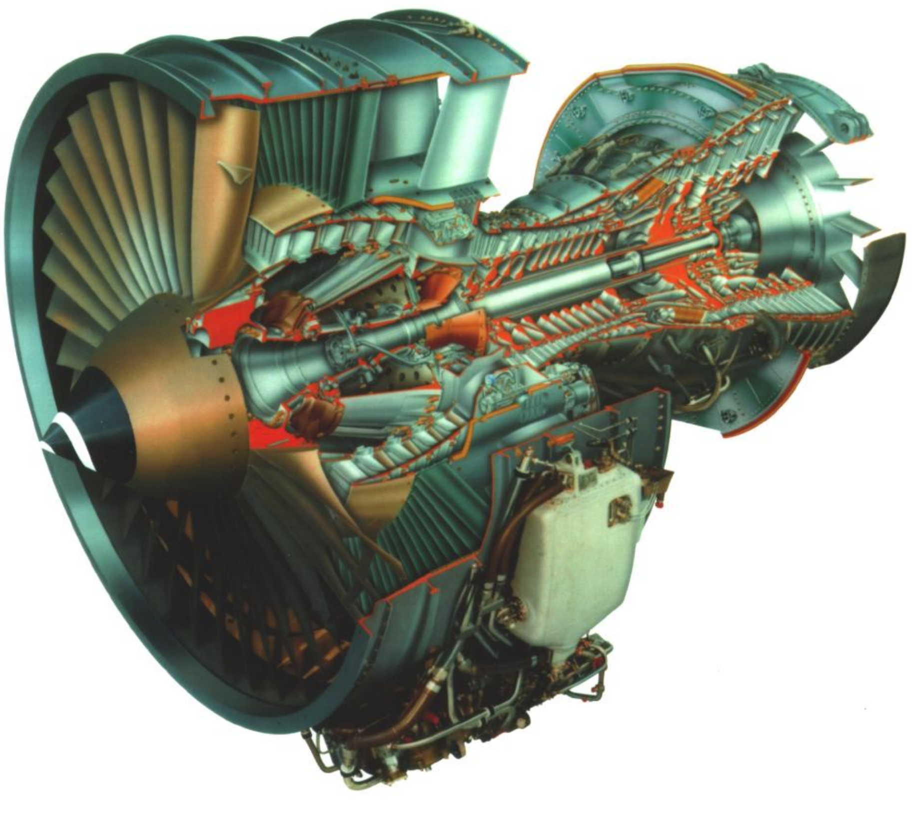
PART OF PHASE C
PRILIMINARY LAYOUT OF TEST FACILITY AND RECOMMENDED TESTS FOR CFM56 ENGINE FAMILY



ГИПРОНИИ АВИАПРОМ

ABBREVIATION

APCS: Automated Process Control Systems

ACS: Automated Control System

OEM: Original Engine Manufacturer

POL: Petroleum, Oil, Lubrication

QEC: Quick Engine Change

SUMMARY

In this phase of CFM56-5C engine test project, the preliminary layout of test cell facility for CFM56 engines family is prepared with considering the received initial data from Final Customer related to engine technical characteristics and available area for construction. The preliminary layout is delivered to Final Customer as #6 drawings of different floors of test facility, its cross sections and also general view of thrust measurment system. This document describes the test facility parts which are illustrated in the drawings attached to this document in more details.

Also, some information related to test procedure of CFM56-5C engine is presented. The information includes some recommended inspections and parameter checking before engine testing, required equipment and materials for testing and the technology of testing based on OEM documents.

1. CHAPTERE ONE:

TEST FACILITY PRELIMINARY LAYOUT

1.1 SITUATIONAL ANALYSIS OF PLAN

In this phase of CFM56-5C engine test project, the preliminary layout of test cell facility for CFM56 engines family is prepared with considering the received initial data from TUGA which required for implementation phases C of the project for the construction, development and operation of CFM56 engine test cells. This initial data categorized in four topics that mentioned below:

Available area for construction

Environmental protection

Climatological information

Engine technical data

1.1.1 Test Bench Situational Position

The proposed area for test cell construction is located in TUGA Company and is surrounded by gas turbine buildings, chemical storage, GYM and dining room. Since TUGA Company is located far enough from residential places, the test cell construction, which is designed considering the acoustic and pollution limits, is located far from residential places with the minimum distance of 400 m and could meet local standard rules.

From the accessibility and transit point of view, the location of test cell should be considered in the place with truck traffic possibility easily for engine and required equipment transportation.

1.1.2 Climatological Considerations

From the pollution point of view, the exhaust system should be far enough from dining room and have enough height. In addition, the dominant wind direction should be considered in order to not ingest hot gases into test cell intake. According to the wind rose diagram showed in Figure 1, the most dominate wind directions which have the possibility of 18% and 10%, are in the band of N-W and W-NW with the maximum possible speed of 21 m/s (exclude the 38.6% of wind in this location which does not have direction). In this regard considering the above mentioned parameters and test cell dimensions, the test cell is located in a way that exhaust system placed far from dining room and in the west of proposed location.

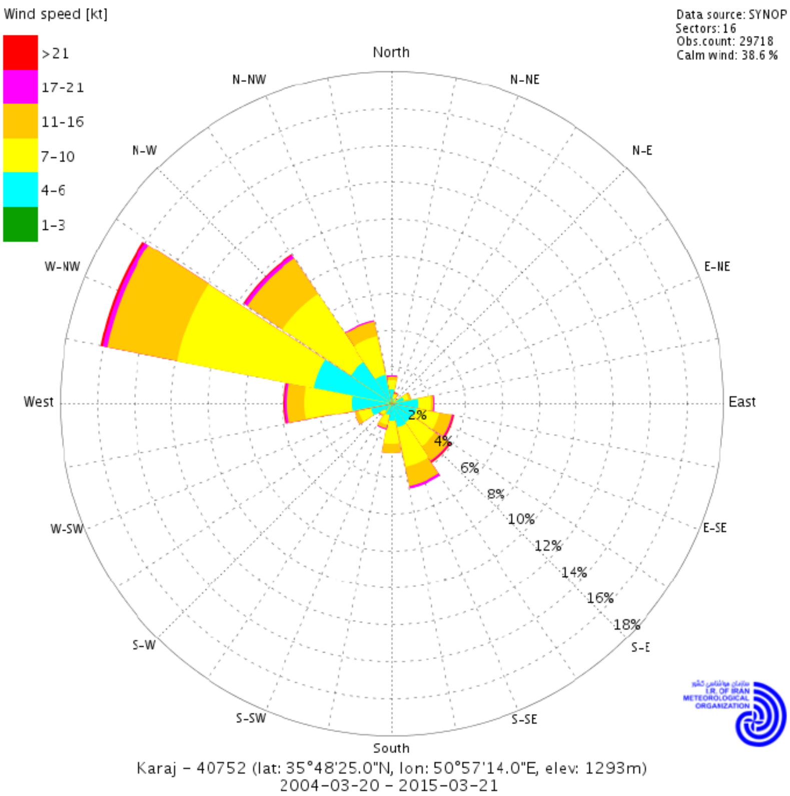


Figure 1: Wind rose Diagram of Karaj

1.1.3 Acoustic Considerations

The maximum noise level outside the test-cell, inside the control-room, in other rooms of the building, in the test chamber where some parts of the engine might deteriorate due to acoustic-fatigue should be specified. In current state, feasibility study of position and location of the test cell is required with respect to the acoustic considerations.

Noise criteria of considered areas in Iran are represented in Table 1 (based on document received from TUGA).

Table 1: Limitations of sound pollution in Iran according to region type

Region Type	Day: 7 AM to 10 PM (dB)	Night: 10 PM to 7 AM (dB)
Industrial Residential	70	60
Industrial	75	65

The acoustic engineering section needs to know the engine power and sound spectrum in order to choose the most efficient noise treatment system and test-cell intake and exhaust devices. Nevertheless, the sound attenuation due to distance is considered here.

The sound waves are radiated in the surroundings and sound energy is extracted by the air molecules. To the attenuation due to the increasing distance (the sound energy is radiated through a spherical surface centered on the noise source and the surface increases with the square of the distance), one must add the attenuation due to "molecular absorption". It is a function of the air temperature and humidity, and of the sound frequency.

Directivity of a typical turbofan engine in the class of CFM56-5 is represented in Figure 2.

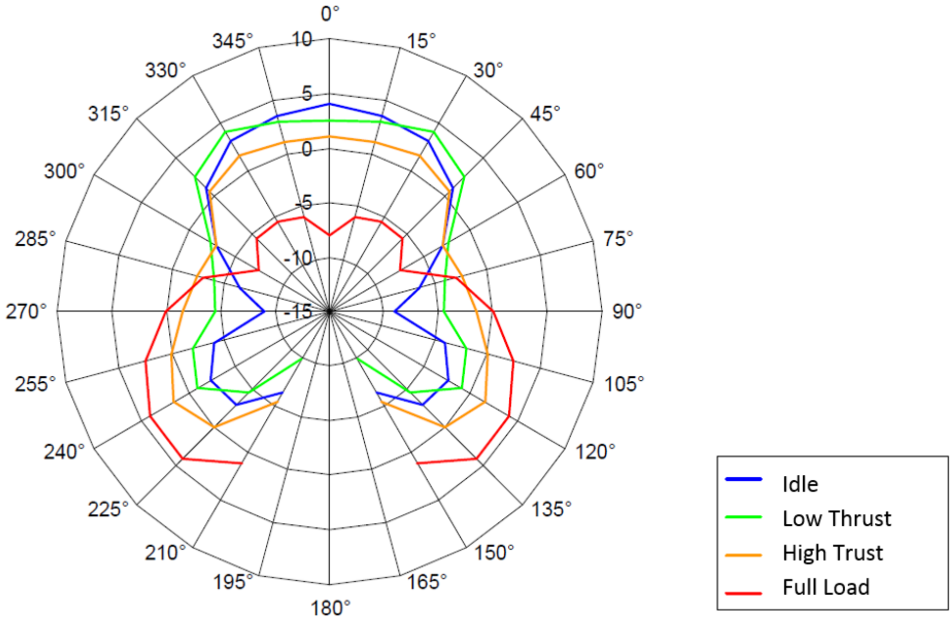


Figure 2: Directivity dependent on various load levels

With mapping this directivity index in the test bench location, we have:

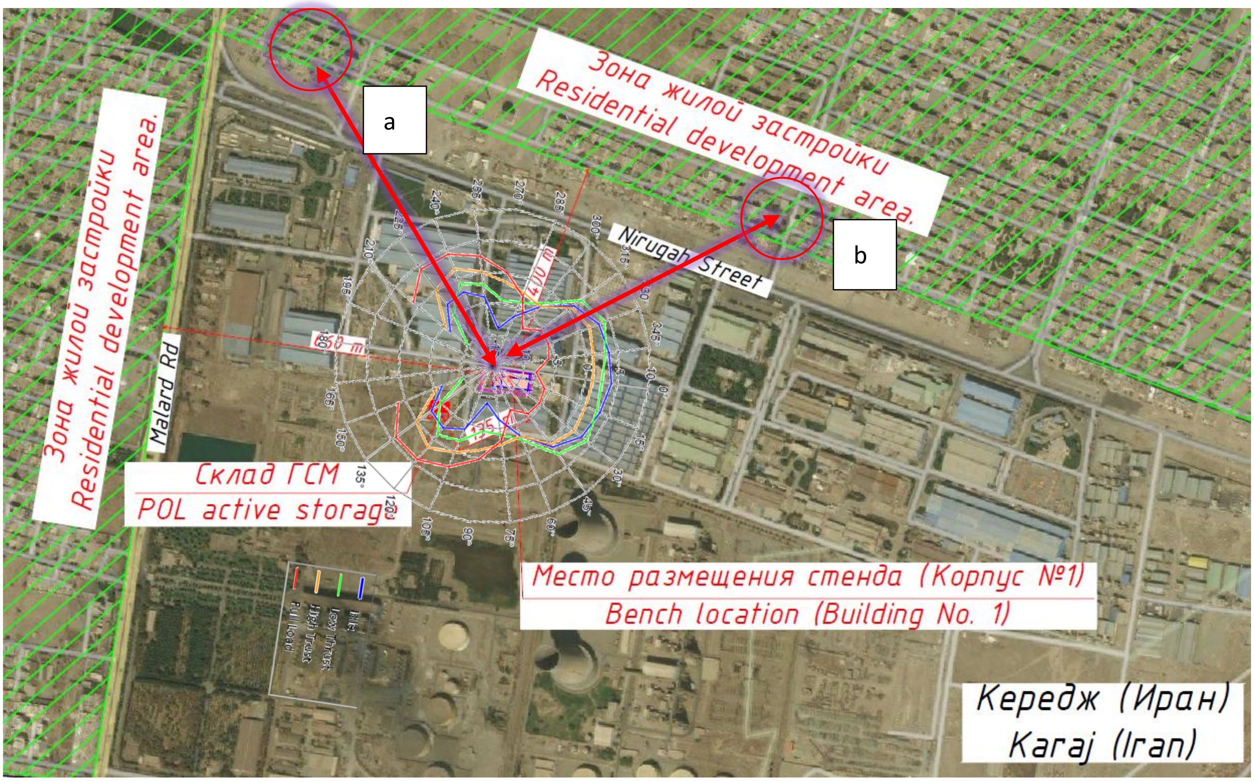


Figure 3: mapping directivity index in the test bench location

a: distance ~ 600m due to exhaust acoustic emission during High thrust rating
b: distance ~ 500 m due to inlet acoustic emission during Low thrust rating

The sound power level (A weighted) of the Airbus family was calculated based on the measured values at a distance of 60 meters (Table 2).

Table 2: Noise emissions during engine text runs

Sound Power Level in dB(A) @60 m				Sound Pressure Level in dB(A) @60 m		One Engine SPL @60m	
Type of Aircraft	Engine Type	Idle	Max power	Idle	Max power	Idle	Max power
A340-200/300	-5C	131.0	152.9	84.4	106.3	81.4	103.3
A321	-5B	131.2	154.1	84.6	107.5	81.6	104.5
A320	-5B	131.1	153.0	84.5	106.4	81.5	103.4
A320	-5A	131.1	152.9	84.5	106.3	81.5	103.3
A319	-5B	130.8	152.2	84.2	105.6	81.2	102.6

From this table, sound pressure level of one engine is calculated at “a” and “b” zones. The results are as follows:

Table 3: Approximated sound pressure level in zone a and b

Sound Power Level in dB(A) @zone “a”			Sound Power Level in dB(A) @zone “b”
Type of Aircraft	Engine Type	Max power	Idle
A321	-5B	84	63

These sound pressure levels are within an acceptable range so that they can be attenuated through test cell walls, baffles, and silencers. These acoustic treatments can be effective as much as at least about 20 decibels. It may be useful to apply sound barrier in the case of excessive acoustic emission.

1.2 PRILIMINARY LAYOUT OVERVIEW

According to proposed area by Final Customer for test facility construction, the preliminary plan which is illustrated in Figure 4 is considered including two different areas: test facility and POL active storage.

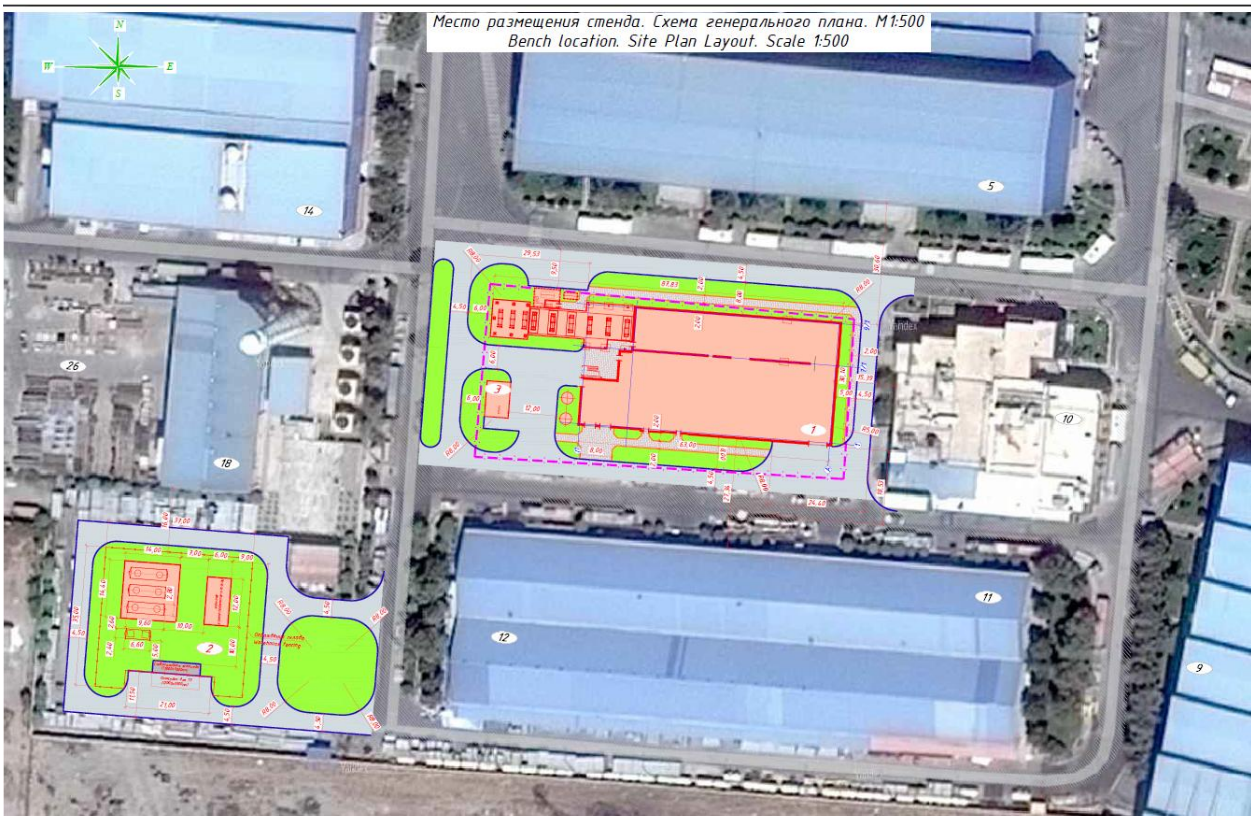


Figure 4. Priliminary plan of test facility

1.2.1 Test Bench Sections and Plans

The preliminary design of the civil part of the test facility is prepared in four floors: ground floor, 1st floor, 2nd floor and 3rd.

- The ground floor layout is presented in Sheet#1.
- The 1st floor layout is presented in Sheet#2.

- The 2nd floor layout is presented in Sheet#3.
- The cross section drawing is presented in Sheet#4.
- The longitudinal section drawing is presented in Sheet#5.

The mentioned drawings represented in the Appendix “A” of this document. Appendix “B” represents the fire hazard standards and tables.

The ground floor rooms are:

Preparation area, test chamber, stairway, corridors, telecommunication room, equipment storage room, switchboard room, air start system compressor room, antechambers, fire extinguisher system rooms, oil room, electric units room, cleaning facilities storage room, water closet, heat supply station.

The 1st floor rooms are:

Preparation area, test chamber, stairway, corridors, air start system compressor room, test data processing room, archives, telecommunication room, switchboard room, control room, APCS room, cleaning equipment storage room, antechamber, men locker, shower room, water closet.

The 2nd floor rooms are:

Preparation area, test chamber, stairway, corridors, production engineering office, process equipment storage room, telecommunication room, switchboard room, cooling system room, air start system process room, fuel valve room airlock, fuel system room airlock, fuel system ventilation chamber airlock, fuel valve room, fuel system ventilation chamber, testing bench fuel system process room, hydraulic pump loading system room, engine air start system valve room, generator loading system process room, exhaust ventilation room, supply ventilation room.

1.2.2 POL Active Storage

This area is designed for fuel loading, pumping, filtering and storage. This area should be located far enough from high technology equipment buildings and work offices and close enough to test cell. In addition, this area should have enough dimensions to store the required fuel for engine testing.

Therefore, the best possible place for locating the fuel storage is selected as shown in presented drawings.

1.3 TEST FACILITY ROOMS

The considered spaces and rooms in the preliminary layout of CFM56 engine test facility are as follows.

Preparation area

The preparation room is the area where Quick Engine Change (QEC) Kit are added to the engine and it is fixed on an adapter and set-up prior to the test in order to minimize the non-running time of the engine inside the test room. The oveahead handling equipment will be provided to handle the engine and adapter between the test chamber and the preparation shop that enters via a door located in a sidewall of the test chamber. As shown in drawings the considered length,width and height for preparation area are approximately 33,20.7 and 10 meters respectively.

Telecommunication room

An environmentally controlled centralized space for telecommunications equipment that usually houses a main or intermediate cross-connect. These rooms are usually located near switchboard rooms. The selection criteria for the location of a telecommunications rooms is the following:

1- Location

Room Obstructions - The area must be free of obstructions that may interfere with the installation of the cable and infrastructure needed to manage the cabling and equipment.

Cable Chases - An area should be designated for chase space.

Ceiling Obstructions - The room's ceiling or the ceiling around the room cannot be obstructed by air conditioning ducts or electrical conduit unless such space has been determined to be allowable.

Cable Distance - The furthest telecommunications outlet must be within an acceptable length.

Electrical Devices/Panels - Electrical devices must be located within a reasonable distance from the selected telecommunications area.

Service Closets - The designated telecommunications area must be near service closets and chases so the cost of extending those services is not excessive.

2- Size

Acceptable Size - Adequate space must be present in the room for racks, UPS power, air-flow, and hardware installation space.

Size Determination - The telecommunications space shall be sized according to the number of outlets serviced by the designated location and the type of networking hardware required including UPS.

Additional Space - Extra space shall be designated for special services such as security and other planned devices.

Distance Around Equipment - A minimum distance is required around telecommunications equipment and racks.

As shown in sheet#1,2 and 3 the considered length,width and height for telecommunications room in each floor are approximately 3,3 and 4 meters respectively.

3- Environment

Lighting - adequate lighting is required around racks and punch down boards.

Air conditioning - The room should have a controlled air environment which does not allow the room to exhibit temperature extremes.

Security - The area designated for a telecommunications room must be able to be secured either with a lock or a card reader system.

4- Electrical Power

Power Circuits- Dedicated power is required for both UPS and non-UPS equipment. Separate circuits shall be installed for both types.

Power Locations - The UPS power circuit must be within six feet of the designated location of the UPS system within the room.

Switchboard room

The switchboards are located in special rooms in each floor. Its considered area is about twice a telecommunication room. As shown in sheet#1 the considered length,width and height for switchboard room in 1st floor are approximately 3,9 and 4 meters respectively. Also, the considered length and width for telecommunications room in 2nd and 3rd floor are approximately 3 and 6 meters respectively (sheet#2 and 3). The height of switchboard room in second floor is 3.95 m and in the third floor is about 3 meter. These rooms are typically required to be fire rated. The switchboard room must also be ventilated to control heat buildup from the equipment.

Complete transformer substation and switchgear station

CCTS is a set of energetic facility that is designed to receive, convert and distribute electrical power to three-phase electricity.

In an electric power system, switchgear is the combination of electrical disconnect switches, fuses or circuit breakers used to control, protect and isolate electrical equipment. Switchgears are used both to de-energize equipment to allow work to be done and to clear faults downstream. This type of equipment is directly linked to the reliability of the electricity supply.

Electric unit room

Electrical devices and power supplies are included in this room. This room is placed next to the test cell on the first floor. As shown in sheet#1 the considered length,width and height for electric unit room are approximately 6,7.5 and 4 meters respectively.

Air start system compressor room

Air start system includes a compressor room, pressurized tanks (air receiver), piping, filters and any other required equipment and air start process room. Typically, the systems can provide both engine start air and shop air for the facility and its systems. Typically features of Air Start Systems include

1- pressurized tanks

They should be sized for multi-start capacity and capacity to “motor” the engine in the event of an emergency. They provide safe storage of air and have local and remote pressure monitoring. As shown in sheet#1 the pressurized tanks are located on the ground floor, adjacent to air start system compressor room.

2- Air Compressors

Single or Dual units of air compressors provide fast recharge rates and housed in weather resistant acoustical enclosures with filters to protect equipment. They may be programmed to provide automatic running to maintain availability, and remote condition/operation monitoring. As shown in sheet#1 the air start system compressor room is located on the ground floor to have better accessibility for transportation planning.

3- Air Dryer

To remove moisture from the air supply

Low maintenance for high operational availability

Pressure Control Valve with remote operation and pressure adjustment

Hose with quick release coupling to engine starter unit

This place which includes air compressor and air pressurized reservoirs should be located far from test cell offices and dining room for safety considerations. On the other hand, this place should be located close enough to engine in order to use as less work as possible for air pumping.

As shown in sheet#1 and 2 the considered area and height for air start system compressor room are approximately 124.2 m^2 and 8.4 meter respectively.

Fire extinguisher system room

A fire detection and protection system has to be installed in the engine room and test stand to protect the test stand and the exposed areas of the engine when it is installed on the test stand. Flame detectors will be provided as the triggering device for the automatic operational mode. The extinguishing agent will spray water containing an anti-freeze additive, and will be distributed around the engine by means of spray rails.

The fire protection systems for the test stand and exposed areas of the engine will be capable of operation for a certain duration at a defined flow rate and pressure at the spray rails. These requirements give a reservoir capacity and a delivery pressure and flow as specified above.

Fire extinguisher system room is placed on the first floor near the test chamber, engine preparation area, oil room, and electric unit room and under the control room.

Discharge of the fire suppressant will be manually trigged by an operator at the control console or automatically by sensing devices located in the protected areas. As shown in sheet#1 the considered length,width and height for fire extinguisher system room are approximately 3,7.5 and 4 meters respectively.

Oil room

It is a support room containing oil preservation systems, oil system unit, hydraulic system unit, fuel preservation systems, and vacuum pump to support engine testing operations. It could be provided skid-mounted storage, pump and filter packages specifically designed to meet the needs of engine facility requirement. Also, flow meters or sensing lines containing fuel or oil shall not be located in the control room and they can be placed in oil room. Oil room is located on the first floor between electric unit room and fire extinguisher system room. As shown in sheet#1 the considered length,width and height for oil room are approximately 8,7.5 and 4 meters respectively.

Preservation systems are considered for filling the fuel and oil system of engine by oil after testing and evacuating the oil from these two systems before testing the engine which does not work for a long time. These units are equipped with tanks, pumps and vacuum pumps and should be located near fire extinguishing system because of safety considerations.

Equipment storage room

A reasonable space is required for some equipment storage. This room may be located on the 1st floor near telecommunication room and switchboard room. As shown in sheet#1 the considered length,width and height for equipment storage room are approximately 3,6 and 4 meters respectively.

Cleaning facilities storage room

There should be considered a room to place cleaning facilities. Such a room is located on the first and second room. As shown in sheet#1 the considered length,width and height for cleaning facilities storage room are approximately 2.1,3 and 4 meters respectively. Also, the considered length,width and height for this room in second floor are approximately 2.2,2.2 and 4 meters respectively (sheet#2).

Test data processing room

This room is configured for process of data captured from engine test cell. This room is on the second floor. As shown in sheet#2 the considered length,width and height for Test data processing room are approximately 5.7,6.4 and 4 meters respectively.

Archives

A proper space should be considered to archive documents and data.

Control room

A control console with instrumentation and computer components used to control and monitor all facility and engine systems relevant to engine testing. This includes an engine control system to control and monitor the engine under test. Management of all systems and operation of the engine under test is carried out with the help of the automated control system, which includes an automated control system (ACS) rack and ACS cabinets with control equipment. As shown in sheet#2 control room placed on the seconed floor and the considered length,width and height for this room are approximately 9,7.5 and 4 meters respectively.

APCS room

APCS develops a process for minimizing human error in plant operations and maintenance.

APCS is a system consisting of personnel and equipment combined with the software used for the automation of functions of personnel management of industrial facilities.

In this case, PLC is implemented as APCS.

Management of all systems and operation of the engine under test is carried out with the help of the automated control system, which includes an Automated control system rack and ACS cabinets with control equipment.

In the APCS, Siemens control equipment, switching equipment and workstations implemented through the systems units are located. The PLC is also connected by the Ethernet to the DACS.

The PLC has three functions described as follow:

1. The first role is to assure the safety, control the fuel pressure, the oil pressure, the exhaust gas temperature, the engine's shaft rotation is also controlled in the PLC
2. The second role is to realize the normal control command of the engine
3. The third role is to realize the control of the remote installation.

As shown in sheet#2 APCS room placed on the seconed floor and the considered length,width and height for this room are approximately 9,7.5 and 4 meters respectively.

Production engineering office, Process equipment storage room

These two places which are located close to each other in 3rd floor are considered for any required manufacturing. As shown in sheet#3 the considered length,width and height for Production engineering office are approximately 6,6 and 3 meters respectively. Also, the considered length,width and height for Process equipment storage room are approximately 4,2.2 and 3 meters respectively.

Heat supply station

Heat supply station provides heating energy required for test cell rooms and equipment. It is placed on the first floor. As shown in sheet#1 the considered length,width and height for heat supply station are approximately 6.4,6.4 and 4 meters respectively.

Supply ventilation room, Cooling system room, Exhaust ventilation room, Fuel system ventilation chamber

These places are considered for ventilation and air conditioning of test facility. Heating and cooling systems shall be arranged to achieve all of the following:

- Reduction of exposure of vital system elements to fire, explosion, and damage by metal
- Elimination of the introduction of ignition sources by components of heating systems
- Minimization of the passage of fire through ductwork
- Elimination of pockets in which flammable vapors can accumulate

Ventilation systems shall be arranged to draw heavier-than-air vapors or gases from near the floor level and discharge them to a safe location. Where lighter-than-air gases are used, similar ventilation shall be provided but arranged to exhaust from ceiling level and with calculations based on the ceiling area. Ventilation for lighter than-air gases shall be designed to prevent pocketing of such gases at ceiling level.

Rotating elements of fans shall be of nonferrous or non-sparking materials, or the casing shall consist of, or be lined with, such material. Where ventilation is provided, each cell or room handling flammable or combustible liquids or flammable gases shall have its own ventilation system to avoid interconnecting multiple hazards.

These rooms are located in 3rd floor. Supply ventilation room and exhaust ventilation room are located close to each other.

As shown in sheet#3 the considered length,width and height for supply ventilation room are approximately 6.7,12 and 3 meters respectively. Also, the considered length,width and height for exhaust ventilation room are approximately 15,7.2 and 3 meters respectively.

The length,width and height which considered for cooling system room are approximately 12,9.2 and 3 meters respectively. The considered area and height for fuel system ventilation chamber are approximately 17.2 m^2 and 3 meter respectively.

Fuel valve room airlock, Fuel system room airlock, Fuel system ventilation chamber airlock

This place is considered at the entrance of fuel rooms in order to not change the air pressure suddenly.

Fuel valve room, Testing bench fuel system process room

These places are considered for measuring and controlling the fuel flow. These rooms are located in 3rd floor. It shoud be noted the considered area and height for, testing bench fuel system process room are approximately 39 m^2 and 3 meter respectively (sheet#3).

1.4 TEST FACILITY STRUCTURES AND ITEMS

The considered structures and items in the preliminary layout of CFM56 engine test facility, in test cell section, are as follows.

Inlet

The inlet of airflow is designed as peripheral open regions located on the top of inlet stack in order to prevent:

- The foreign object, rain and snow entrance into the test cell.
- The re-ingestion engine combustion products emitted from the exhaust. (This factor reduces the influence of wind direction on test cell performance.)

Typically, air intakes are vertical or horizontal. The balance is between a smoother and more uniform air flow to the engine with a horizontal intake because the flow is directly to the engine without having to turn through 90°, or fewer ground effects and less dust ingested with the vertical intake which can be several meters above ground level. Additionally, because the airflow from a horizontal intake to the engine can be strongly influenced by external wind behaviour it is usual to shield such intakes from cross winds by building side walls in front of the test chamber. Unfortunately, this can help generate eddy phenomena in the airflows close to the inlet; just the type of flow distortion the inlet should be designed to avoid.

On the other hands the design of the vertical intake can take advantage of the 90° turn in the air flow to largely eliminate shear effects due to wind direction.

Horizontal intakes are usually less likely than vertical intakes to re-ingest engine combustion products emitted from the exhaust. Not only must the physical separation of inlet and exhaust be adequate, it is also important to avoid placing the exhaust stack upwind of the inlet, particularly if the inlet is to be vertical.

Within the constraints of the site, to avoid wind shear effects in the intake region the orientation of the test cell should be such that the intake is aligned with the prevailing wind direction. This will minimize but not eliminate the problem. The vertical intake has been chosen for the new test bed currently being used by the manufacturer because recent advances in the design of flow splitters and straighteners have substantially reduced the adverse effects.

Wind effects and pressure losses across the inlet section could affect the engine performance and possibly damage engine components. The drop in total pressure across the inlet system could result in lower pressure inside the cell. An excessive depression in cell pressure could affect the air flow stability around the engine and accuracy of measurements, and might possibly cause structural problems. The depression is due to the dynamic pressure loss resulting from the velocity in the test-bay and from the flow pressure losses in the air intake duct (flow open area, bends, acoustic panels, turning vane and screens).

It is generally recommended to limit the depression to a maximum of 150 mmH2O. The reason is not the structural load limit but the correction on the measured parameters and the operation of the engine in conditions similar to those existing in open air test stand. At this limit the engine will be working in conditions close to those of free air, with minimum corrections necessary during measurements and their correlations.

Non-turbulent intake air is crucial to proper, reliable aero engine testing. A proper inlet system must be designed in conjunction with the exhaust of the facility to achieve total cell flow with minimal cell depression. The distortion depends on the duct shape of the air inlet stack, the flow dividing obstacles and their profiles (acoustic panels), the shape and number of bends, the number and profile of the turning vanes, the wire mesh and bird duct screens, the flow length between the downstream edge of the panels or airflow straighteners and the engine inlet where vortexes and wakes are damped.

Inlet air flow rotary vanes

This item which is known turning vanes has been used in order to rotate the flow from vertical stack into horizontal test chamber as less distortion and pressure drop as possible. These turning vanes always leave a wake in the airflow. These wakes are regions of lower pressure which not only contribute to the overall depression in cell pressure but also cause an increase in the overall turbulence level of the air entering the engine. The design of these vanes to minimize these adverse effects is a significant part of inlet design.

Air filter netting

This item has been used in order to prevent the ingestion of birds, dust and any foreign objects into the engine. Also, this can play a role in decreasing the flow distortion. Sizing of screens is a compromise: too small a mesh will produce an unacceptably high pressure loss, whilst too large a mesh will allow potentially damaging objects to pass into the test cell.

Sound suppressing baffles

The silencer baffles are required to reduce engine noise emitted from the intake to acceptable levels add considerably to aerodynamic losses in the inlet region, because the restricted width of the channels necessary to produce sound attenuation introduces additional friction to the airflow. In this regard, the maximum velocity through the most restricted portion is 50-55 m/s.

The intake is acoustically lined in one to three rows of splitters. The needed noise attenuation quality is dependent on the thickness of the splitters and the distance between them. The larger distance and thickness are used to attenuate low frequency noise whereas high frequency noise is best attenuated by small distance and thickness. Generally, the dimensions of the silencer depend upon:

- The required total air mass flow
- The maximum cell depression
- The detailed design of the splitters.

Also, this equipment has been used to provide a uniform air flow at the chamber room which is where the engine performance measurement is taken.

Shutter gates electric drive with gearbox

Test chamber room was equipped with the shutter gate in order to protect against ingestion of dust and any foreign objects when the engine is shut dawn. The shutter gate operates electrically with using electric drive and gearbox.

The electric drive and gearbox should be robust, durable and reliable, while at the same time work quietly .The roller shutter motors move the shutters smoothly and precisely. It have an integrated slow speed, which guarantees a gentle stop travel. This protects the roller shutters and reduces noise build-up significantly.

Testing chamber

Test chamber is an enclosed space prepared and well equipped for engine test plans and such activities as, measuring components performance quantities, altitude condition simulation, and engine monitoring. A perfect test chamber would contain nothing to cause airflow distortion. It would have frictionless surfaces and contain idealised air flows with no viscous effects and no vortex formation or exhaust gas re-ingestion. Frictionless flow is not possible, but careful design can prevent ingestion of exhaust gases and minimise vortex formation.

Once the air in the chamber room is accelerated into the engine, wall-flow separation may occur if the cell remaining air flow is not high enough to prevent low flow velocities at the cell walls. Also, since air is drawn from all the directions nearby of the engine inlet, the components of the velocities may cancel beneath it: the flow angular velocity and the stagnation point originated are a perfect opportunity for vortices to form. This phenomenon depends on air and engine speed and its height above the floor. It is known that bigger engines lead to a lower peripheral air flow available thus increasing the risk of vortex formation. Some important parameter that must be considered in design of test chamber are mentioned below:

- Aspect ratio of the test chamber,
- mass flow of air through the engine,
- cell bypass ratio

Aspect ratio of the test cell is the ratio of the cross section area of the chamber to the intake area of the engine.

When testing in an enclosed sea-level test bed, the ejector action of the jet stream from the engine exhaust as it enters the augmenter (exhaust collector) induces a flow of secondary air through the cell. The secondary air flow through the intake stack and the test chamber and the temperature is very close to the ambient temperature outside the test-bench. This secondary flow rises aerodynamic effects which causes forces to act on the thrust frame which increases errors in the measurement of thrust.

The three main effects are:

- A force on the inlet bell-mouth due to the air entering predominantly from the forward direction;

- A force on the nozzle due to the reduced external static pressure caused by the increased velocity of the secondary air as it approaches the exhaust collector;
- A drag force on the framework supporting the engine due to the secondary air flowing over it.

The biggest effect is in the region of the engine nozzle exhaust because here the air velocities are highest as the air is drawn into the augmenter.

Increasing of a secondary air flow will generate a more static pressure gradient between engine inlet and nozzle exhaust; this will require a larger thrust correction factor (TCF), which is undesirable. To minimize the correction to be applied to measured values, these secondary air speeds should be kept as low as possible. Generally, this area will be of a sufficient cross section so that the air velocity approaching the engine inlet will not exceed approximately 15 meter per second.

As shown in drawings the considered dimensions for test chamber cross section is 10 m* 10 m and the test chamber lenght is 39 m. The lenght of test cell is restricted by acoustic and aerodynamic considerations. By decreasing in test cell lenght specially in augmenter section, the thrust correlation factor increase and the thrust measurement will encounter high amount of error. As shown in the cross section drawing the thickness of the roof in superstructure interface is 1.5m and in another parts of the roof is 0.5m. the roof is not integrated and this two parts of the roof are seprated .

<u>Overhead handling equipment (crane)</u>

The oveahead handling equipment will be provided to handle the engine and adapter between the test chamber and the preparation shop that enters via a door located in a sidewall of the test chamber. The crane is capable of transporting a fan engine, adapter, and test nacelle. Generally the monorail enters the cell perpendicular to the wall, and turns 90° before placing the engine in a thrust stand.

Lifting maintenance platform

The engine under test is mounted from overhead positioned on cell centerline. The purpose of the lifting maintenance platform is to raise mechanics and their equipment to a convenient working height to perform checks and routine maintenance on the engine.

Overhead maintenance platform

Overhead maintenance platform has been designed in order to a safe, stable work surface for the operators.

Thrust measurement system

The thrust stand assembly will consist of the upper (fixed) assembly that interfaces with the superstructure and the lower (floating) assembly that is suspended from the upper assembly by flexure plates. The bottom of the lower assembly provides a neutral docking frame interface to accommodate engine test adapters. The lower assembly incorporates an adapter locking system. Two load cells are mounted on both sides of the thrust frame for measuring engine thrust. To minimize the effect of ambient temperature changes on load cell accuracy, a temperature-controlled heating jacket is provided for each load cell. The thrust frame is designed with sufficient stiffness so the weight of the docking frame and engine has minimal influence on the thrust reading from any load cell. The drawing of the thrust stand is presented in appendixA.

Augmenter shutter

This item has been used in order to protect the engine and test chamber from dust and any foreign object which coming from exhaust side.

Rolling shutter augmenter gate

A rollering shutter is sectional overhead door consisting of many horizontal slats hinged together. The door is raised to open it and lowered to close it and the action is motorized. It provides protection against ingestion of dust and any foreign objects.

Augmenter

The hot exhaust gases entrain the peripheral flow that surrounds the engine; both streams enter the augmenter which then acts as an ejector pump.Augmenter tubes serve two purposes: to cool the exhaust gases by mixing them with fresh air, and to attenuate noise. High energy exhaust gases flow at high velocity while the secondary airflow has low energy and velocity. The mixing and interaction between these two airflows at the entrance to the augmenter produce an ejector pump effect which is both important and quite complex. However, the ejector effect will have an effect on the pressure at the engine inlet possibly altering its performance.

The pumping efficiency depends on the augmenter geometry:

- $\frac{L}{D}$: length/diameter ratio
- $\frac{A_n}{A_{aug}}$: Ratio of the nozzle exhaust cross-section to augmenter tube cross section.
- $\frac{X}{D_n}$: Ratio of the distance between engine nozzle exhaust and augmenter inlet to nozzle diameter.

At indoor sea level thrust test bed cell bypass ratio is influenced by the distance between engine nozzle exhaust and augmenter inlet, ratio of the diameters of the nozzle and augmenter tube and primary flow pressure and temperature.

The mixing of the two streams reduces the temperature and velocity of the primary flow which are critical factors in the reduction of both cell cost and noise pollution. The noise of a jet of air is directly related to the velocity of the jet hence reducing the velocity of the jet will reduce noise pollution generated by the test cell. The reduction of exhaust jet temperature reduces the need for complex designs and use of heat resistant materials in the exhaust system, thus reducing cost (and size) of the test cell.

Exhaust stack

The exhaust stack acts as a large chimney at the end of the exhaust ductwork, expelling diluted and cooled combustion gases into the atmosphere.

Exhaust stack be lined with acoustic absorbent material to reduce engine noise emissions. The cross-sectional area of the stack determines the velocity of the exhaust gases and a compromise must be reached between having as low a velocity as possible to minimize aerodynamic noise generation pollution within the silencer while having a sufficiently high velocity to carry the exhaust gases into the atmosphere .

Exhaust silencer

Exhaust silencer is used to protect against excessive noise levels depending on

the cell size and the type of engines being tested. Acoustic absorbents used in exhaust systems have an upper temperature limit of around 400ºC. So it is necessary to introduce a significant degree of cooling either using water cooling or by having a large secondary air flow. The secondary air flow plays a very important role in cooling of engine exhaust gas flow. In order to improve the efficiency of the anti-noise treatment and control the back pressure, the velocity has to be limited at 30-40 m/s along the exhaust stack.

Oil and fuel spillages collection system

This system has been designed to transfer oil and fuel spillage to the collection tank that located underground.

Underground tank

Underground tank is including connected underground pipes that is used to contain fuel. Fuel and oil spillage tank should be located far from the test chamber with the minimum distance of 15 m.

<u>Oil and fuel spillage collection tank</u>

This tank has been used in order to collect fuel and oil spillage.

2. CHAPTERE TWO:

TEST PROCEDURE

2.1 INTRODUCTION

This chapter gives some information which is required for testing the engine and contains the general data that follows:

- Tests CFMI recommends after module replacements.
- Procedures for inspections before you start the tests.
- The types of data you must keep records of during the tests.
- Engine motoring procedure.
- Engine start procedure.
- Power setting procedure.
- Engine shutdown procedure.

The other recommended engine tests are:

- Operating Limits: Limits for components and engine parameters.
- Functional Test: Engine break-in procedures (also known as run-in, generally is the procedure of conditioning a new piece of equipment by giving it an initial period of running usually under light load but sometimes under heavy load or normal load.) and functional checks for the FADEC system, vibration limits, and operational response.
- Acceptance Test: Certified thrust limits and general performance calculations.
- Fan Trim Balance: Fan trim balance procedures.

2.2 CFMI RECOMMENDED TESTS FOR MODULE REPLACEMENT.

The following table presents the recommended test after module replacements in CFM56-5C engine.

Table 4. Recommended test after module replacement

Module	Functional Test	Acceptance Test
Fan Major Module	Yes	Yes
Core Engine Major Module	Yes	Yes
LPT Major Module	Yes	Yes
Accessory Drive Major Module		Yes
Fan and Booster Module	Yes	Yes
No. 1 & 2 Brg Support Module	Yes	Yes
Compressor Rotor Module	Yes	Yes
Compressor Front Stator Module	Yes	Yes
Compressor Rear Stator Module	Yes	Yes
Combustion Chamber Module	Yes	Yes
HPT Nozzle Module	Yes	Yes
HPT Rotor Module	Yes	Yes
HPT Shroud/Stg 1 LPT	Yes	Yes
Nozzle Module	Yes	Yes
Inlet Gearbox & No. 3 Bearing Module	Yes	Yes
Transfer Gearbox Module	Yes	
Accessory Gearbox Module		Yes

2.3 PRE-TEST INSPECTION

This is a sample inspection checklist. It is important to carefully make an inspection of the test facility and engine before each test. Make sure all procedures necessary to prepare for each test are complete before you start the tests.

The recommended procedure for pre-test inspection is:

1) Read the engine log or cards. Make sure all necessary inspections have been made on work that is complete. Make sure the engine has been released for test.
2) Make sure the engine lubrication system is correctly serviced.
3) Make sure the starter is correctly serviced.
4) Make sure the fuel supply is the correct type, and there is an adequate quantity for the test.
5) Make a check of the basic engine piping as follows:
   (1) Make sure the piping is correctly installed.
   (2) Set the fuel boost pumps to ON.
   (3) Examine the fuel piping connections for leaks.
   (4) Set the fuel boost pumps to OFF.
6) Make sure the basic engine rigging is correct.
7) Make a visual inspection of VSV linkages for correct installation and serviceable condition.
8) Make sure the VBV doors and flex shafts are serviceable.
9) When you operate engines without accessories on the customer pads, make sure the plugs are installed on the gear shafts. Examine the engine and test cell thrust and support mounts before each engine run to prevent injuries and damage to the engine and test cell.

10) Make sure the test cell thrust and support mounts are serviceable as follows:

(1) Examine the mounts for cracks.

(2) Make sure the mounts are tightly attached.

(3) Make sure the mounts are installed correctly.

11) Make sure the drain hoses are correctly installed.

12) Make sure the starter air pipes and leads are correctly installed.

13) Make sure all the electrical connections are tight.

14) Make sure all vibration leads, sensors, and instruments are correctly installed.

15) Make sure the throttle operates correctly.

16) Make sure the shut-off solenoid valve on the HMU is OFF.

17) Set the ignition switch to OFF (AUTO Mode position).

18) Examine the fan and core inlets and exhaust for unwanted materials.

19) Make sure the test cell floor is clean.

20) Start the electrical power supply for the engine, test cell, and panel.

2.4 RECORDED DATA

In this section, the list of data that should be recorded during engine operation and test is presented.

1) When you operate the engine, monitor it closely. Keep a record of all important events and of these parameters:
    - T2 (fan inlet front frame hub section temperature)
    - Fuel Flow
    - Time

- N1 and N2 speeds
- Oil pressure
- Exhaust Gas Temperature (EGT).

2) When you do functional checks, such as a start, monitor and keep a record of these parameters:
    - Time necessary for the engine to get to idle Peak EGT
    - Fuel flow during start
    - EGT (at all speeds except minimum idle)
    - T2 (fan inlet front frame hub section temperature)
    - The conditions for the starter air supply
    - Oil pressure
    - N1 and N2 speeds
    - Vibration indications (which include amplitude and frequency).

3) When you do the performance points for steady-state operation, keep a record of the indications from all of the instruments. Do not include the indications from the instruments that measure transient conditions.

4) Keep a record of this minimum data on the test log sheets for each engine installation:
    - Date.
    - Serial number of the engine.
    - Number of the test cell.
    - The type of oil used in the engine.

- The serial numbers of the major pieces of test equipment (bell-mouth, test cowlings, etc.)
- Fan nozzle area.
- Primary nozzle area.
- The lower heating value (LHV) of the fuel during the last analysis.
- The specific gravity of the fuel.
- The temperature of the fuel at the flow meter.
- Fuel and oil leakage as specified (fuel quantity and oil quantity, measured separately).
- Data from starts:
  - T12 (fan inlet temperature).
  - Maximum EGT.
  - The pressure and temperature of the starter air.
  - The time necessary for the engine to get to minimum idle after you energize the starter.
  - The fuel flow during start and the peak fuel flow.
- Barometric pressure, adjusted to 32°F (0°C).
- Absolute humidity (in grains).
- Engine operation time.
- Time of day when you did test.
- Transient thrust data:
  - Maximum N1 and N2 speeds.
  - Maximum EGT.
  - The time necessary for the engine to get to 95 percent of take-off thrust.

- The time necessary for the engine to get to maximum EGT.
- Specific comments about the problems that occur during the tests.
- Readings of all engine instruments at each steady-state data point.
- CFMI recommends you make a record of all ARINC output during the tests. The ARINC output will help you do troubleshooting if you have engine problems.

2.5 ENGINE MOTORING PROCEDURE

It should be noted that do not motor the engine without a positive fuel inlet pressure. The fuel pump and HMU are lubricated by fuel and damage to these components can occur.

A. Dry-Motor the engine as follows:

(1) Set the Throttle Lever to IDLE.

(2) Set the Mode Selector to NORMAL.

(3) Set the Master Lever to OFF.

(4) Set the electrical power to ON and wait for the engine parameters to come on the screen.

(5) Set the Manual Start Switch to ON.

(6) Set the Mode Selector to CRANK.

(7) Permit the engine to become stable at approximately 4000 rpm N2 speed. And, monitor the oil pressure for a positive indication.

(8) Obey the starter limits.

The complete procedure for motoring check which should be performed before engine testing is described in Functional Test section.

2.6 ENGINE START PROCEDURE

There is an automatic and a manual start procedure.

2.6.1 Automatic Start Procedure

It should be noted that dry-motor the engine for the specified time to prevent engine damage.

1) If the engine was operated during the previous 2 hours, dry-motor the engine for approximately 90 seconds. Perform an additional 90-second motoring if the EGT is greater than 100°C refer to section 2.5.
2) Do this checklist before you start the engine:

    (1) Set the Master Lever to OFF.

    (2) Set the Throttle to IDLE.

    (3) Set the Mode Selector to NORMAL.

    (4) Set the electrical power to ON.

    (5) Make sure a sufficient fuel supply is available.

    (6) Set the Fuel Boost Pump to ON.

    (7) Make sure there is sufficient fuel pressure.

    Caution: carefully monitor the start cycle when you use lower starter inlet air-pressure than we recommend. Low air pressures at the starter inlet can cause higher peak EGT and unusual lengths of time to get to idle.

    (8) Make sure there is sufficient starter inlet air pressure. CFMI recommends 25 psig (172 kPa) with the start valve open.

    (9) Make sure the engine oil quantity is correct.

Caution: do not operate the engine above minimum idle with the cowl open to prevent damage to the cowls.

3) Start the engine as follows:

    (1) Set the Mode Selector to IGNITION.

(2) Set the Master Lever to ON and carefully monitor these indications:

NOTE: The ignition system automatically starts at 16 percent N2. The fuel system automatically starts at 22 percent N2. The starter and ignition systems automatically stop at approximately 50 percent N2.

Caution: do not try to start the engine without an N1 or N2 indication. Engine damage can occur.

a) Monitor the N1 and N2 speeds until the engine becomes stable at minimum idle and:
    a. Stop the start cycle if there is no N1 or N2 indication before the engine gets to idle
    b. Stop the start cycle if the N2 decays or stops before the engine gets to idle.
b) Carefully monitor the fuel flow until the engine becomes stable at minimum idle.
c) Carefully monitor the EGT until the engine becomes stable at minimum idle. The engine will usually light off approximately 10 seconds after the start cycle starts, and 2-3 seconds after the fuel flow starts.
d) If a normal light-off occurs, go to step 8.
e) If the engine fails to light-off, go to step 7.
f) If the engine stalls, continue as follows:
    a. If the ECU senses a stall, the fuel will be shut off, and the engine will dry motor for 7 seconds.
    b. The ECU will lean-out the fuel 7 percent, and after 7 seconds will attempt another start.

c. If the engine stalls again, the fuel will be leaned-out another 7 percent, and another attempt will be made. If the engine start is not successful, the start will be aborted. In this event, set the Master Lever to OFF, and set the Mode Selector to NORMAL.

d. Do troubleshooting to correct the problem. Another start attempt can be done after the starter has cooled.

g) If the engine fails to light-off, continue as follows:

a. If the light-off does not occur in 10 seconds (15 seconds in very cold weather), the ECU automatically de-energizes the ignition system. It also stops the fuel flow and dry-motors the engine for 30 seconds.

b. The ECU will automatically energize both igniters. At 30 seconds, the FADEC system automatically starts the fuel flow again.

c. If there is no light-off in 10 seconds (15 seconds in very cold weather) during the second start cycle, the ECU will automatically de-energize the ignition system, stop the fuel flow and de-energize the starter.

d. The ECU system automatically dry-motors the engine again for 30 seconds, and gives a START ABORT fault message.

e. If the ECU cannot automatically start the engine, set the Master Lever to OFF.

f. Set the Mode Selector to NORMAL.

g. Do troubleshooting to correct the problem. Do the automatic or manual start procedure again after the starter is cool.

h) Carefully monitor the oil pressure until the engine becomes stable at minimum idle.

   a. You must get an indication of oil pressure before the engine becomes stable at minimum idle.

   b. High oil pressure can occur in cold weather because of high oil viscosity. When the temperature of the oil increases, the oil pressure decreases.

   Caution: obey the oil pressure limits. If the engine operates at low oil pressures, bearing damage can occur.

   c. If the oil pressure is not in the limits after the oil temperature becomes stable, stop the engine and do troubleshooting.

2.6.2 Manual Start Procedure

It should be noted that dry-motor the engine for the specified time to prevent engine damage.

1) If the engine was operated during the previous 2 hours, dry-motor the engine for approximately 90 seconds. Perform an additional 90-second motoring if the EGT is greater than 100°C refer to section 2.5.
2) Do this checklist before you start the engine:

   (1) Set the Master Lever to OFF.

   (2) Set the Throttle to IDLE.

   (3) Set the Mode Selector to NORMAL.

   (4) Set the electrical power to ON.

   (5) Make sure a sufficient fuel supply is available.

   (6) Set the Fuel Boost Pump to ON.

(7) Make sure there is sufficient fuel pressure.

Caution: carefully monitor the start cycle when you use lower starter inlet air-pressure than we recommend. low air pressure at the starter inlet can cause higher peak egt and unusual lengths of time to get to idle.

(8) Make sure there is sufficient starter inlet air pressure. CFMI recommends 25 psig (172 kPa) with the start valve open.

(9) Make sure the engine oil quantity is correct.

Caution: do not operate the engine above minimum idle with the cowl open to prevent damage to the cowls.

3) Start the engine as follows:

(1) Set the manual start switch to ON.

(2) Set the Mode Selector to IGNITION.

(3) Set the Master Lever to ON when the engine gets to 20 percent N2.

(4) Move the Mode Selector to NORMAL when the engine gets to idle.

(5) Monitor the fuel flow indications until the engine is stable at min idle.

(6) Monitor the EGT indications until the engine becomes stable at min idle.

(7) Monitor the oil pressure until the engine becomes stable at min idle.

a. You must get an indication of oil pressure before the engine becomes stable at minimum idle.

b. High oil pressure can occur in cold weather because of high oil viscosity. When the temperature of the oil increases, the oil pressure decreases.

Caution: obey the oil pressure limits. if the engine operates at low oil pressures, bearing damage can occur.

c. If the oil pressure is not in the limits after the oil temperature becomes stable, stop the engine and do troubleshooting.

2.7 POWER SETTING PROCEDURE

A. Set all power settings to the N1 or as limited by EGT.

B. Move the throttle slowly and smoothly when you set the power. Move the throttle quickly only when it is specified in a procedure.

C. Let the engine become stable at idle power:

(1) After an engine start, let the engine be stable at minimum idle for a minimum of 5 minutes before trying to operate at a higher power setting.

(2) For any restart after any engine shutdown (including trim balance), stabilize the engine at minimum idle for a minimum of 10 minutes followed by a 2-minute accel to the last point of test.

(3) After a deceleration from higher power to minimum idle, let the engine be stable at minimum idle for a minimum of 5 minutes before you return to higher power.

(4) Prior to any non-emergency engine shutdown, minimum idle for 10 minutes minimum.

D. Use a minimum of one minute to move the throttle from minimum idle to maximum continuous power or higher unless given different instructions.

E. Use these procedures when you move the throttle during decelerations:

(1) Use a minimum of one minute to move the throttle when you decelerate to minimum idle from stable operation (more than 20 seconds) at or above maximum continuous power.

(2) You can make a short (one second) deceleration to minimum idle from nonstabilized operation (not greater than 20 seconds) at conditions up to take-off power.

2.8 ENGINE SHUTDOWN PROCEDURE

A. If you used high power settings, operate the engine at minimum idle for 10 minutes minimum to permit the engine to become cool.

B. Set these controls and monitor these parameters:

(1) Set the Throttle Lever to IDLE.

(2) Set the Master Lever to OFF.

(3) Set the Mode Selector to NORMAL.

(4) Set the Manual Start Switch to OFF.

Caution: make sure the egt constantly decreases to prevent damage to the engine.

(5) Carefully monitor the EGT. If it does not decrease (or if you get a tailpipe fire) motor the engine until the fire stops.

(6) Monitor the N2 speed.

(7) Monitor the fuel flow indication to make sure the fuel flow stops.

(8) Set the fuel supply to OFF.

Caution: dry-motor the engine for the specified time to prevent engine damage.

(9) Dry-motor for a minimum of 2 minutes after the core stops or until the EGT is less than 100 °C. This is necessary to reduce the possibility of a core seizure that can lead to heavy blade and seal rubs.

2.9 OPERATING LIMITS

The limits in this procedure are applicable for all engine tests. The operator must know all of these limits before engine operation and testing.

2.9.1 Limits

A. Starter limits:

Requirements: 4 consecutive cycles, 2 minutes each, maximum.

Remarks: 20 seconds of no operation between cycles to permit the starter rotor to be lubricated again.

Remarks: After 4 cycles, wait 15 minutes to permit the starter to cool before a new start try or before motoring.

Remarks: Do not engage the starter when the core speed is more than 20 percent.

B. Engine Start Time:

(1) Time to light-off

Remarks: Light-off should occur in 2 to 3 seconds after 22 percent N2 is reached during an auto start, or in 2 to 3 seconds after master lever is turned on during a manual start. If light-off does not occur within 10 seconds, place fuel shutoff lever to OFF, continue to motor engine with starter for 60 seconds, then de-energize starter.

(2) Start to minimum idle.

Requirements: Corrected Start Time (CST) must be less than Specification Start Time (SST).

Remarks: Do a troubleshooting if the CST is more than the SST.

C. Ignition System:

(1) Duty cycle

Requirements: Continuous use is permitted.

Remarks: Use one system first, and then the other system. Make a minimum of one start on each system during the test schedule. The maximum ignition unit environmental temperature must not be more than 250°F (121°C) with continuous ignition.

D. Rotors Do Not Rotate:

(1) Fan (N1): Core engine operation with fan rotor or low pressure turbine (LPT) rotor seized.

Requirements: Maximum permitted core engine speed is the maximum motoring speed.

Remarks: Do not operate the engine without an N1 rotation indication. Set the master lever to OFF and the mode selector to NORMAL. Do not try to start the engine. Do troubleshooting to find the cause.

(2) Core (N2): Core seized because of seal or shroud interference, or starter shaft is sheared.

Remarks: Do not start engine with the starter if core rotor is seized. Do not attempt to start the engine until the core rotor turns freely by hand. Attempts to turn a seized rotor using the starter can cause heavy seal rubs and a loss in performance.

Remarks: If the engine is cold, troubleshoot the starter.

E. Lubrication System:

Caution: shut down the engine if there is any oil pressure change of 5 psi (34 kpa) during steady-state operation. do troubleshooting to find the cause. engine damage can occur if the engine is operated with this defect.

(1) Oil Pressure

NOTE: The corrected oil pressure equals oil supply pressure (lube pump discharge) minus the sump vent pressure plus the oil correction factor.

(a) During start

Requirements: Positive indication before 2000 rpm (N2).

(b) At stabilized minimum idle

Requirements: Type 1 oil: 13 psid (90 kPa) min. Indicated.

Remarks: If not in limits, examine and correct the cause before operation at higher speeds.

Requirements: Type 2 oil: 13 psid (90 kPa) min. indicated

Remarks: If not in limits, examine and correct the cause before operation at higher speeds.

(c) Above approach idle

Requirements: Type 1 oil: 42.3-61.6 psid (292-425 kPa) corrected.

Remarks: If not in limits, examine and correct the cause.

Requirements: Type 2 oil 45.7-64.5 psid (315-445 kPa) corrected.

Remarks: If not in limits, examine and correct the cause.

(2) Oil Temperature

(a) At lube pump discharge

Requirements: 285°F (141°C) maximum.

Remarks: The normal oil supply temperature to the engine is 150-285°F (66 to 141°C).

(b) Scavenge oil

Requirements: 320°F (160 °C) maximum.

Remarks: Decrease the power until the temperature is in limits if it is more than 320°F (160°C) for 5 minutes. Shut down the engine and examine it to find the cause. High scavenge oil temperature can be an indication of lube system problems.

(c) Transient operation

Requirements: 347°F (175°C) maximum.

(3) Oil consumption

Requirements: 0.14 gal/hr (0,5 l/hr)

Remarks: This limit is necessary to make sure the engine will not use more than the operational consumption rate when it is installed.

F. Fuel System:

(1) Fuel flow

Requirements: 400-680 pph (182-308 kg/h) during start.

Remarks: Fuel flow more than 680 pph (308 kg/h) can cause you to go above the EGT limits.

(2) Fuel inlet

Requirements: 10-50 psig (69-345 kPa gage).

Remarks: At the inlet to the fuel pump during all test modes to include transients.

(3) Facility flow verification

Requirements: Both facility fuel flow meters must not be more than 0.5 percent different.

G. Under-Cowl Air Temperature Limits:

(1) All test operations

Requirements: 401°F (205°C) maximum forward of station 221 and 500°F (260°C) maximum aft of station 221.

Remarks: You must monitor under-cowl temperatures during break-in and all test operations. If temperatures are not in limits, examine the engine for air leaks. Correct all air leaks and continue the test run.

H. Exhaust Gas Temperature (EGT):

(1) Start

Requirements: 1337°F (725°C) maximum.

(2) All other operations

Requirements: Not more than the determined limits in engine manual.

I. Vibration Levels:

NOTE: Blade tip and seal clearances are sensitive to vibrations. Careful attention should be given to vibration levels to prevent taking heavy rubs.Transient: Applies during all engine accel/decel (throttle movements).Steady State: Applies after 1 minute of stabilized operation.

(1) Fan/LPT Rotor (N1)

NOTE: N1 vibrations up to 8.0 mils are allowed during mechanical checkout. Do not operate the engine if the LP vibration levels exceed 8.0 mils DA.

NOTE: To minimize impact on performance, do not exceed the following vibration levels during testing. Initiate trim balance if limits are exceed:Transient - LP/REV vib peaks up to 4.5 mils DA are serviceable during accel/decel. LP/REV vib peaks from 4.5 to 6.0 mils DA are serviceable if the steady-state limit (4.0 mils DA) can be met after one minute stabilization at the point where the peaks occur.Steady State - 4.0 mils DA after one minute of operation. Send engine to the shop for module repair or re-balance if not within limits for transient acceptance after fan trim balance procedure.

(a)

Requirements: Transient:

Remarks: LP/REV vib peaks up to 4.5 mils D.A. are serviceable during accel/decel. LP/REV vib peaks from 4.5 to 6.0 mils D.A. are serviceable if the steady-state limit (4.0 mils D.A.) can be met after one minute stabilization at the point where the peaks occur.

(b)

Requirements: Steady State:

Remarks: There is a steady-state limit of 4.0 mils D.A. after one minute of operation.

Remarks: Send the engine to the shop for module repair or to be balanced again if it is not in the limits for transient acceptance after the fan trim balance procedure.

(2) Core Engine Rotor (N2)

Requirements: according to determined limits in engine manual.

Remarks: The HP/REV vibration levels are for the No. 1 bearing vertical and turbine rear frame sensors. Send the engine to the shop if you go over the 1.6 ips acceptance limit.

J. Oil Leakage:

(1) Starter pad

Requirements: Maximum 10 ml/h.

Remarks: Replace the magnetic seal or sealol seal.

(2) AGB IDG pad

Requirements: Maximum 10 ml/h.

Remarks: Replace the magnetic seal or sealol seal.

(3) Rear hydraulic pump pad

Requirements: Maximum 10 ml/h.

Remarks: Replace the magnetic seal or sealol seal.

(4) Fuel pump pad

Requirements: Maximum 10 ml/h.

Remarks: Replace the magnetic seal or sealol seal.

(5) Forward sump

Requirements: Maximum 30 ml/h.

Remarks: Repair the engine.

K. Fuel Leakage:

(1) Fuel manifold shroud

Requirements: No leaks are permitted.

Remarks: Replace the defective component.

(2) VSV actuator

Requirements: Maximum 120 ml/h.

Remarks: Replace the defective component.

(3) VBV motor

Requirements: Maximum 120 ml/h.

Remarks: Replace the motor.

(4) Fuel pump at drive pad

Requirements: Maximum 120 ml/h.

Remarks: Replace the pump.

(5) Heat exchanger

Requirements: Maximum 20 ml/h.

Remarks: Replace the heat exchanger.

(6) Hydro-mechanical Unit (HMU)

Requirements: Maximum 180 ml/h.

Remarks: Replace the HMU.

(7) HPT clearance control valve

Requirements: Maximum 180 ml/h.

Remarks: Replace the HPT clearance control valve.

(8) LPT clearance control valve

Requirements: Maximum 180 ml/h.

Remarks: Replace the LPT control valve.

(9) Burner staging valve

Requirements: Maximum 180 ml/h.

Remarks: Replace the burner staging valve.

(10)Rotor Active Clearance Control (RACC valve)

Requirements: Maximum 180 ml/h.

Remarks: Replace the RACC valve.

L. Idle Stabilization Time:

(1) After a start

Requirements: 5 minutes minimum

Remarks: Do an engine shutdown procedure immediately if an emergency occurs refer to section 2.8. It is not necessary to permit the engine to become stable first. Engage the starter at 2000 N2 rpm and motor the engine to equally cool it if the engine was operated at high power settings. Do not motor the engine if it will cause more damage.

(2) For a restart after any engine shutdown (including trim balance)

Requirements: 10 minutes minimum followed by a 2 minute accel to the last point of test.

(3) After operation at more than 2100 N1 rpm before you accelerate again

Requirements: 5 minutes minimum.

(4) Before a non-emergency shutdown

Requirements: 10 minutes mininum.

M. Rotor Speeds.

(1) High pressure rotor (N2):

(a) Minimum idle.

Requirements: 8500 ±25 RPM

(b) Maximum physical speed

Requirements: 15183 rpm (100 percent = 14460 rpm).

Remarks: Do an overspeed inspection if your speed is more than this limit.

(2) Low pressure rotor (N1):

(a) Maximum physical speed

Requirements: 4798 rm (100 percent = 4784 rpm).

Remarks: Repair the cause if your speed is more than this limit.

(3) Low pressure rotor (N1):

(a) Maximum physical speed

Requirements: 4985 rm (100 percent = 4784 rpm).

Remarks: Repair the cause if your speed is more than this limit.

N. Variable Stator Vane (VSV) schedule

Requirements: Monitor ARINC Label 370 Bit 13. Read on the ARINC bus.

Remarks: If Bit 13 is set, the VSV did not track correctly at the time. Do troubleshooting of the VSV system

O. Variable Bleed Valve (VBV) schedule

Requirements: Monitor ARINC Label 370 Bit 12. Read on the ARINC bus

Remarks: If Bit 12 is set, the VBV did not track correctly at that time. Do troubleshooting of the VBV system

P. High Pressure Compressor (HPC) Discharge Pressure:

(1) Minimum Idle.

Requirements: 25±1 psig (159 ±6,9 kPa gage) or 8500 ±25 N2 rpm.

(2) Takeoff

Requirements: 520 psia (3585 kPaA) maximum.

Q. Ratings:

(1) Takeoff power setting

Remarks: Maximum of 5 minutes

(2) Maximum continuous

Remarks: Continuous

R. Nozzle Areas (Test):

(1) Primary Nozzle Area

Requirements: 2390.6 sq in. (15424 sq cm) ±1.5 percent

Remarks: Ambient.

S. Nozzle Areas (Test):

(1) Nozzle Exit Plane Area

Requirements: 2411.8 sq in. (15561 sq cm) ±1.5 percent

Remarks: Ambient.

T. ECU Temperature

Requirements: 221°F (105°C) maximum.

Remarks: Remove power to the ECU when the engine is not in operation.

U. Fault Monitoring

Requirements: Monitor channels A and B of the ECU ARINC output bus.

Remarks: Correct the faults before you continue the test.

V. Dry-motoring for engine cooling:

(1) Start

Requirements: Dry motor for approximately 90 seconds before start if the engine was operated during the previous 2 hours. Perform an additional 90 second motoring if the EGT is greater than 100°C.

Remarks: Obey the starter limits.

(2) Shutdown

Requirements: Dry motor for a minimum of 2 minutes after the core stops or until the EGT is less than 100°C.

NOTE: This is necessary to reduce the possibility of a core seizure that can lead to heavy blade and seal rubs.

2.9.2 ECU Faults

Check and consider the ECU messages and investigate the error causes and fix them considering the ECU message discription presented in CFM56-5C engine manual.

2.10 FUNCTIONAL TEST

The functional check gives instructions to make checks of the engine for:

- Fuel and oil leaks
- Vibration
- Idle speed
- Variable Stator Vane (VSV) tracking
- Variable Bleed Valve (VBV) tracking
- Full Authority Digital Electronic Control (FADEC) operation
- other mechanical functions.

It is important to carefully make an inspection of the test facility and engine before each test. So before you start the functional check do some pre-test inspections which are mentioned below:

- Do the pre-test inspection refer to section 2.9 for the correct limits. Monitor all of the parameters for changes during the test procedure. You must do this if the parameters are in normal limits or not. A change in the parameters can be an indication of a problem.
- Make sure the engine is correctly prepared for operation. Refer to section 2.3.

- Do a pre-test inspection refer to section 2.4 before you operate the engine.

2.10.1 Leak Inspection

A. If necessary, open the cowling doors.

B. Start the fuel supply to the engine. Set the fuel supply to 10-50 psig (69-345 kPa gage).

C. Examine all fuel components and tubes for leaks. If leaks are found, stop the fuel supply and correct them. Do the leak inspection again after you correct the leaks.

D. If the leak inspection is complete, close the cowling doors.

2.10.2 Motoring Check

It shoud be noted you must do this check before you start the engine according to the procedure described below.

A. If necessary, open the cowling doors.

B. Motor the engine refer to section 2.6.

C. Let the engine be stable at the maximum motoring rpm and monitor these items:

(1) Make sure there is a positive indication of oil pressure.

(2) Make sure the N2 indication is approximately 4000 rpm.

(3) Make sure there is an N1 indication.

(4) Make sure the VSV are closed (38.7 ±1 degree).

(5) Make sure the VBV are open (more than 36 degrees).

(6) Make sure there is no leakage from the oil system tubes or components.

D. If necessary, start the fuel supply to the engine.

Caution: do not try to modulate the fuel flow with the master lever. high fuel flows will put too much fuel into the engine.

E. Make sure there is an indication of fuel flow as follows:

(1) Set the Master Lever to ON for a maximum of 15 seconds.

(2) Monitor the fuel flow indicator for a positive indication. If the fuel flow indication is more than 680 pph (308 kg/h), immediately set the Master Lever to OFF.

(3) Set the Master Lever to OFF. When you do this, the fuel flow indication must decrease to 0.

F. Continue to dry-motor the engine for 60 seconds to remove the fuel from the combustor.Refer to section 2.5.

G. Set the Mode Selector to NORMAL.

H. Listen for unusual noises from the engine during the coastdown period.

2.10.3 Start Tests

A. Do an automatic start procedure. Refer to section 2.6.1.

B. Let the engine become stable at minimum idle and make an inspection for leaks or other unusual conditions.

C. If you find problems, do an engine shutdown procedure and correct them before you continue with the test. Refer to section 2.8.

D. Make a check of the minimum-to-approach idle transition as follows:

(1) Set the throttle to APPROACH IDLE and make sure the transition is smooth.

(2) Set the throttle to MINIMUM IDLE and make sure the transition is smooth.

E. Do an engine shutdown procedure. Refer to section 2.8.

F. Fill the oil tank to full.

Note: After filling the oil tank dry-motor (refer to section 2.5 the engine for 3-5 minutes to remove the air from the lubrication system. Look for leaks while you motor the engine.

G. Do a manual start procedure. Refer to section 2.6.2.

H. Operate the engine at or near minimum idle before you accelerate to a higher power setting.

I. If necessary, do an engine shutdown procedure. Refer to section 2.8.

2.10.4 Fadec System Test

A. Perform an engine shutdown procedure. Refer to section 2.8.

B. Remove power to both channels of the ECU (off for at least 30 seconds).

C. Energize both channels of the ECU.

NOTE: One channel will become active, the other channel will be in stand-by. Assuming equivalent health on both channels, the channel in control will alternate on each engine shutdown.

D. Make sure that there are no FADEC faults indicated on the ARINC output. If a fault is generated, perform necessary troubleshooting and clear the fault.

E. Do an engine start procedure. Refer to section 2.6.

F. Let the engine become stable at minimum idle.

G. Make sure of the integrity of the FADEC System as follows:

(1) Slowly accelerate the engine to 11,100 rpm N2, and allow it to become stable.

(2) Decelerate the engine to minimum idle. Let the engine become stable for 5 minutes.

(3) If no faults were generated, continue to the next step. If a fault is generated, perform necessary trouble-shooting to clear the fault, and repeat this test.

H. Do an engine shutdown procedure. Refer to section 2.8.

I. Make sure that the stand-by channel (from the first time through the test) becomes active. This is to make sure that the FADEC system test is done on both channels of the ECU.

J. Repeat steps 2.10.4.D. through 2.10.4.H.

2.10.5 Functional Check Break-In, Vibration Signature, And Transient Checks

It should be mentioned that you can do these checks at the same time.This procedure conditions the surfaces of these seals to help keep clearance deterioration to a minimum in normal service:

- Low Pressure Turbine (LPT) shrouds
- LPT interstage seals
- Compressor Discharge Seals
- Balance Piston Seals.

Carefully monitor the fan vibration levels during all accelerations and decelerations. Do a fan trim balance procedure if the fan vibrations are not in the limits (Max 0.004 in pick to pick). Refer to section 2.11. Also, carefully monitor the core vibration levels during all accelerations and decelerations. Send the engine to the shop if the core vibrations are not in the limits.

Do the checks as follows:

NOTE: Set the engine speeds for the power settings in this test to 10 rpm according to power setting data mentioned in engine manual.

(1) Do an engine start procedure. Refer to section 2.6.

(2) Keep the engine at minimum idle for a minimum of 5 minutes, and write the indications after 5 minutes.

(3) Slowly accelerate (30 seconds) to 3500 rpm N1, hold for 5 seconds, then decelerate (10 seconds) to 3000 rpm N1.

(4) Stay at 3000 rpm N1 for 3 minutes, and write the indications.

(5) Slowly accelerate (30 seconds) to 4200 rpm N1, hold for 5 seconds, then decelerate (10 seconds) to 3800 rpm N1.

(6) Stay at 3800 rpm N1 for 3 minutes, and write the indications.

(7) Slowly accelerate (30 seconds) to 4462 rpm N1. Hold for 5 seconds. Decelerate (10 seconds) to 4100 rpm N1.

(7) Slowly accelerate (30 seconds) to 4443 rpm N1. Hold for 5 seconds. Decelerate (10 seconds) to 4100 rpm N1.

(7) Slowly accelerate (30 seconds) to 4551 rpm N1. Hold for 5 seconds. Decelerate (10 seconds) to 4100 rpm N1.

(7) Slowly accelerate (30 seconds) to 4532 rpm N1. Hold for 5 seconds. Decelerate (10 seconds) to 4100 rpm N1.

(7) Slowly accelerate (30 seconds) to 4684 rpm N1. Hold for 5 seconds. Decelerate (10 seconds) to 4190 rpm N1

(8) Stay at 4100 rpm N1 for 3 minutes, and write the indications.

(8) Stay at 4190 rpm N1 for 3 minutes, and write the indications.

(9) Slowly accelerate (30 seconds) to 4462 rpm N1. Hold for 5 seconds. Decelerate (10 seconds) to maximum continuous (4292 rpm N1).

(9) Slowly accelerate (30 seconds) to 4443 rpm N1. Hold for 5 seconds. Decelerate (10 seconds) to maximum continuous (4273 rpm N1).

(9) Slowly accelerate (30 seconds) to 4551 rpm N1. Hold for 5 seconds. Decelerate (10 seconds) to maximum continuous (4292 rpm N1).

(9) Slowly accelerate (30 seconds) to 4532 rpm N1. Hold for 5 seconds. Decelerate (10 seconds) to maximum continuous (4337 rpm N1).

(9) Slowly accelerate (30 seconds) to 4684 rpm N1. Hold for 5 seconds. Decelerate (10 seconds) to maximum continuous (4398 rpm N1).

(10) Stay at 4292 rpm N1 for 5 minutes, and write the indications.

(10) Stay at 4273 rpm N1 for 5 minutes, and write the indications.

(10) Stay at 4337 rpm N1 for 5 minutes, and write the indications.

(10) Stay at 4398 rpm N1 for 5 minutes, and write the indications.

(11) Slowly decelerate (2 minutes) to approach idle.

(12) Stay at approach idle for 10 minutes, and write the indications.

(13) Slowly decelerate (30 seconds) to minimum idle.

(14) Stay at minimum idle for 5 minutes, and write the indications on the third minute.

(15) Slowly decelerate (30 seconds) to minimum idle.

(15) Review data obtained and compare limits. Refer to section 2.9.If vibration data is beyond limits at any speed, return to that speed, write additional vibration data, and make sure of the results.

Caution: the fan must be balanced prior to proceeding to the next step. Refer to section 2.11.

(16) Slowly accelerate (2 minutes) to 4462 rpm N1. Hold for 30-40 seconds. Slowly decelerate (2 minutes) to minimum idle power.

Caution: the fan must be balanced prior to proceeding to the next step. Refer to section 2.11.

(16) Slowly accelerate (2 minutes) to 4443 rpm N1. Hold for 30-40 seconds. Slowly decelerate (2 minutes) to minimum idle power.

Caution: the fan must be balanced prior to proceeding to the next step. Refer to section 2.11.

(16) Slowly accelerate (2 minutes) to 4551 rpm N1. Hold for 30-40 seconds. Slowly decelerate (2 minutes) to minimum idle power.

Caution: the fan must be balanced prior to proceeding to the next step. Refer to section 2.11.

(16) Slowly accelerate (2 minutes) to 4532 rpm N1. Hold for 30-40 seconds. Slowly decelerate (2 minutes) to minimum idle power.

Caution: the fan must be balanced prior to proceeding to the next step. Refer to section 2.11.

(16) Slowly accelerate (2 minutes) to 4684 rpm N1. Hold for 30-40 seconds. Slowly decelerate (2 minutes) to minimum idle power.

(17) Stay at minimum idle for 5 minutes, and write the indications on the third minute.

(18) Make sure the minimum idle speed is correct.

Do an engine shutdown procedure. Refer to section 2.8. It should be noted you can continue without an engine shutdown if more tests are scheduled.Finally examine all filters, screens, and magnetic chip detectors.

2.11 ACCEPTANCE TEST

This procedure gives the instructions for the Engine Acceptance Test. During the test you will operate the engine at different power settings (fan speeds) for the ambient conditions and collect performance data. You will then compare the performance data to the engine's certified performance levels.

Characteristics of operation for N2, EPR, and WF are provided to help you control engine quality. You can also use this data to do troubleshooting if the performance of the engine is marginal. But, do not use the characteristics of operation to accept or reject an engine. During this test, consider the operating limits that are mentioned before. Refer to section 2.92.9.

A. This test makes sure the indicated EGT and N2 are in the performance limits. It also makes sure that when you set the N1, the engine produces the correct thrust at the applicable power setting.

B. Monitor the fuel flow (WF) and exhaust gas temperature (EGT) carefully. These parameters can help find possible problems and control quality.

C. Do the tests as follows:

(1) Start the engine.

(2) Record the corrected time to MI and peak EGT. Set N1 for the power settings in this test to 5 rpm.

(3) Do the following warm-up procedure:

NOTE: Repeat the warm-up procedure each time the engine is shut down during the test.

(a) Keep the engine at MI for 5 minutes.

(b) Slowly accelerate (one minute) to MC (4292 rpm N1), and keep the engine at MC for 5 minutes.

(c) Slowly decelerate (one minute) to MI, and keep the engine at MI for 5 minutes. Record the indications at 3 minutes.

(4) Record the oil quantity indication at 5 minutes.

NOTE: The electronic oil level sensor automatically compensates for temperature. Do not make manual corrections.

NOTE: Physical fan speed can be monitored on label 045 "NIACTSEL."

(5) Determine the TO physical fan speed.

(6) Slowly accelerate to the physical fan speed calculated in step (5) and set throttle stop at that position.

(7) Do a slow deceleration to 15 percent TO power. Keep the engine at 15 percent TO for 5 minutes.

(8) Quickly accelerate (one second) to throttle stop calculated in step (6) to produce a stable takeoff N1, and at the same time record the elapsed acceleration time to 95% maximum TO thrust. Keep the engine at take off N1 for 30 seconds, then do a quick deceleration (one second) to MI. Keep the engine at MI for 5 minutes.

(a) The elapsed acceleration time must not be more than 5 seconds. If acceleration time is greater than 5 seconds, do troubleshooting for the cause.

NOTE: The following steady state data points are necessary to complete the acceptance test requirements. Conduct these data points with a fan speed modifier level of zero.

(9) Accelerate to MC. Keep the engine at MC for 10 minutes to prepare for the acceptance test.

(10)Record the indications after the engine becomes stable.

(11)Accelerate to TO power. Keep the engine at TO power for 5 minutes and record the indications.

NOTE: It is recommended not to exceed 5 minutes at TO power.

(12)Decelerate to MC. Keep the engine at MC for a minimum of 5 minutes. Record the indications after the engine becomes stable.

(13)Slowly decelerate (2 minutes) to AI. Keep the engine at AI for 5 minutes. Record the indications after the engine becomes stable.

(14)Slowly decelerate MI. Keep the engine at MI for 5 minutes. Record the indications after the engine becomes stable.

(15)Record the oil quantity indication at 5 minutes.

(16)Subtract oil quantity recorded in step (14) from oil quantity recorded in step (4).

(17)If the engine used more than the oil limits, do an oil consumption run.

NOTE: The engine oil consumption obtained by calculation with oil level indications must be considered for information only, as the tolerance of the indicator can be equal to the oil consumption limit.

(18)If oil quantity increased, examine fuel/oil heat exchanger and servo fuel heater for fuel in the oil system.

(19)Do an engine shutdown procedure. Refer to section 2.8.

(20)Select a fan speed modifier (FMN1) as follows:

NOTE: The purpose of the FMN1 is to trim the excess thrust to the required minimum thrust level.

(a) Using TO thrust margin, select the FMN1 level

(b) Install FMN1 plug onto the engine, and do an electronic test.

(c) Calculate engine thrust, EGT, and N2, using both modified and unmodified N1 per calculations in step E(3). In the performance summary, include the FMN1 level and the performance margins with the modifier, the performance margins at the zero modifier level.

E. Performance Troubleshooting.

(1) If the engine fails any performance test point, examine the test data for instrumentation faults, and then repeat the test point, if necessary.

The following Figure shows the recommended acceptance test schedule for CFM56-5C2.

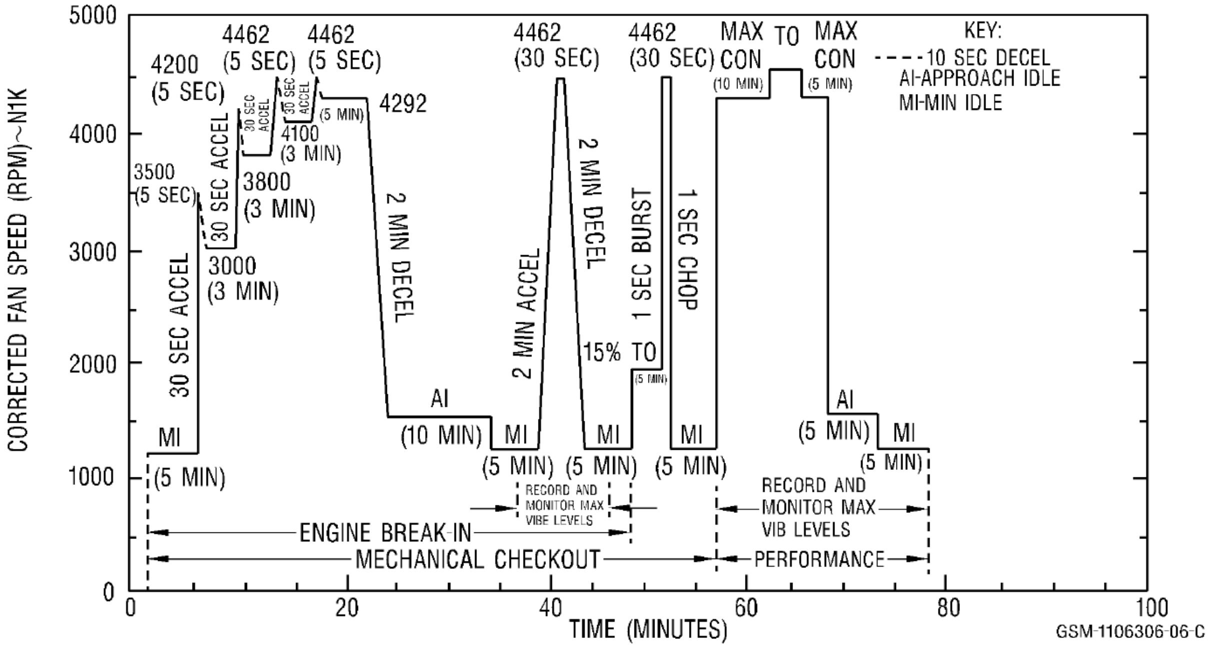


Figure 5: Acceptance test schedule for CFM56-5C2

2.12 FAN TRIM BALANCE

This procedure should be performed before starting the engine test.

Trim balance is done when the No. 1 bearing or Fan Frame Compressor Casing (FFCC) vibration level is more than 4 Mils double amplitude (D.A). It is made to balance the fan rotor and put No. 1 bearing and FFCC levels to 4 Mils D.A. or less for all engine operating speeds.

The test bench aids will give an imbalance indication filtered for frequency of one per revolution of the low pressure rotor and a phase angle of the imbalance location which can be identified to a fixed position on the rotor. This permits to find the correct position to install correction weights when you use one of the procedures which follow.

a) One-shot procedure:

For this procedure it is necessary to have a predetermined matrix which gives the relation of the phase angle to the balance screw locations (phase Lag) and the relation of the vibration amplitude to the balance weight (sensitivity) at specific rotor speeds. It is not possible to give a same matrix for all applications. All engines have different responses which give unique sensitivity relationship. With this procedure, you will find the solution for most of the balance problems and you will have to do it only one time.

b) Vectorial procedure:

The vectorial procedure makes you compare the phase angle and imbalance magnitude of the first run with a subsequent run with added correction weights. Graphical vectorial analysis is used to find accurate phase lag and sensitivity of the engine/analyzer used. A final correct location and amount of weight necessary to trim balance is then calculated.

For these 2 procedures the data which follow are necessary:

1- A vibration signal from the No. 1 bearing vibration sensor with a sensitivity of 100 pc/ g.

2- A vibration signal from the FFCC vibration sensor with a sensitivity of 50 pc/g.

3- A speed signal from the N1 speed sensor. The speed sensor get input from the fan rotation with a 30-tooth wheel attached to the fan rotor. This wheel has one tooth thicker than the others, so that each time it goes in front of the sensor, a unique signal is transmitted. This unique signal is used to find a phase angle in relation to the imbalance indication.

4- A speed signal from the N2 speed sensor which can be filtered to find the origin of the vibrations caused by the HP rotor system.

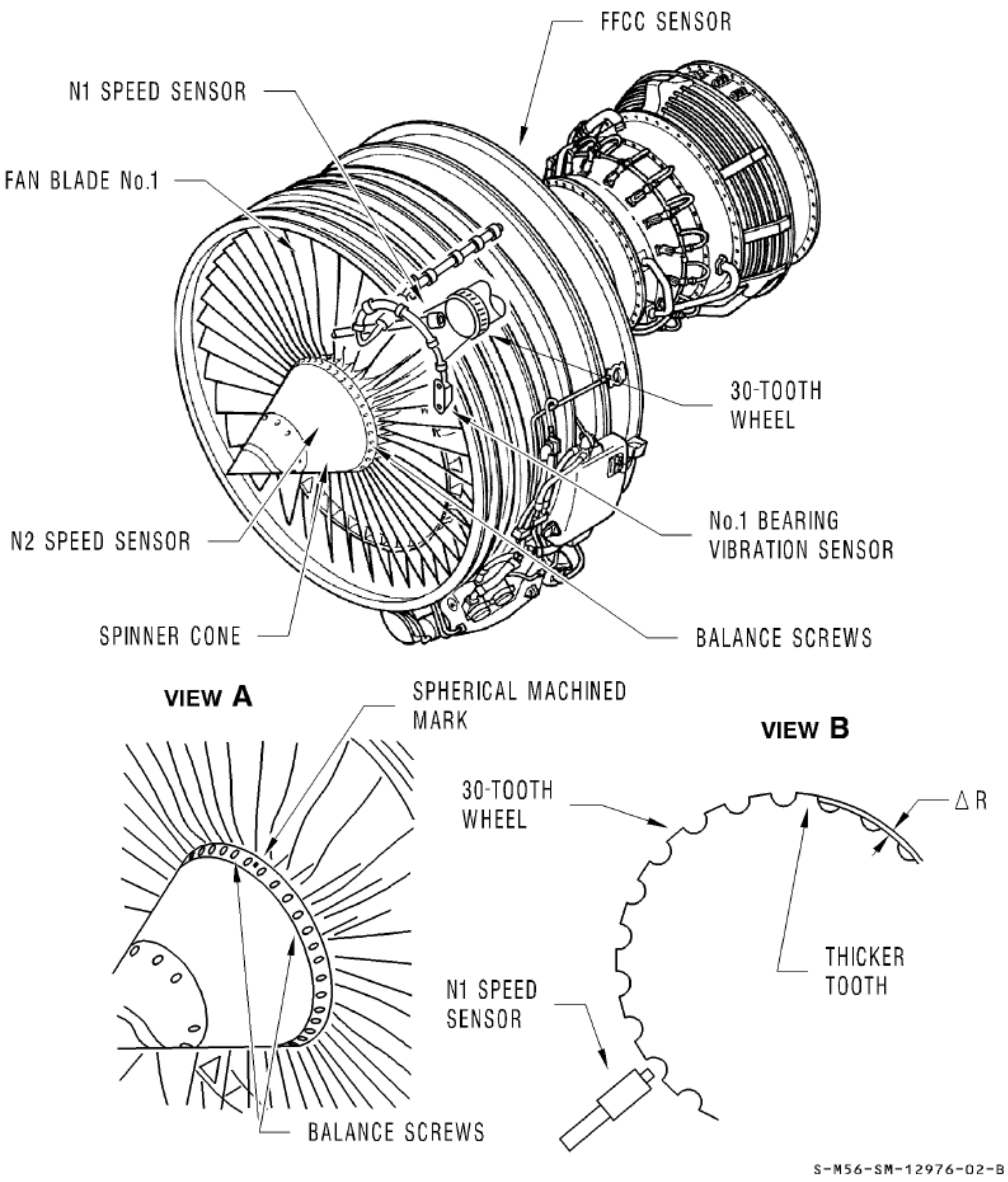


Figure 6: Sensors and Angular References Used for Trim Balance

2.12.1 Definition Of Terms And Symbols

Symbols:

U - Vibration Amplitude.

A - Phase Angle as read on Vibration Analyzer.

W - Balance Correction Weight.

R - Resultant - The result of the vector analysis. (The vector is made when you connect the tip of vector No. 1 to tip of vector No. 2).

L - Location - Screw Location where correction weight is to be installed (centered).

S - Sensitivity - Ratio of an imbalance correction weight (g.cm) by the resultant (Mils) given by this imbalance correction weight, for a given speed and sensor.

NOTE: Subscripts may be added to the below symbols to show the run number.

Terms:

(1) Displacement vector.

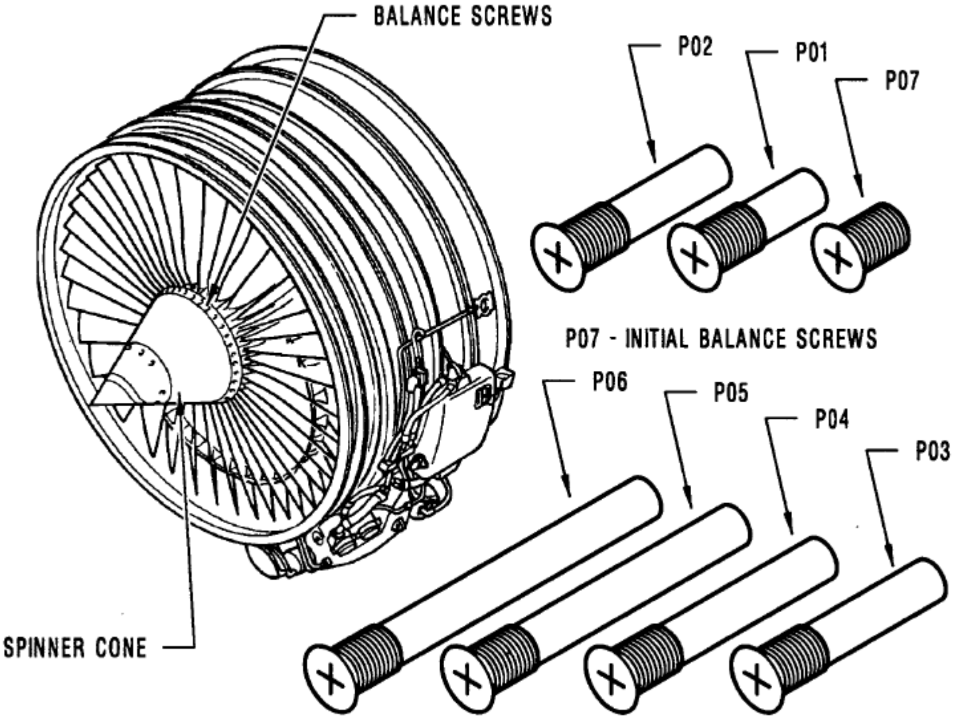
Displacement vector (U) indicated on the balance analyzer is expressed in Mils Double Amplitude (DA). The angular direction is given by the phase angle (A).

(2) Balance screw. Refer to Figure 7.

Balance screws are screws of different lengths installed radially on the spinner rear cone.

(3) Angular orientation.

The angular orientation in relation to the fan is given by a signal from a tooth on the 30-tooth wheel. This tooth has a radius larger than the radius of the others and agrees with the fan reference blade. This blade is indicated by a spherical mark machined on the spinner rear cone.



BALANCE SCREW

P0# MARKED ON SOCKET HEAD

THREADS

TOTAL LENGTH

PO #	TOTAL LENGTH IN.	TOTAL LENGTH MM	WEIGHT Grams	M.W.* GR-CM
P07	0.8	20.3	6.6	0
P01	1.3	33.0	10.0	89.3
P02	1.8	45.7	13.4	174.3
P03	2.3	58.4	16.8	257.5
P04	2.8	71.1	20.2	336.2
P05	3.3	83.8	23.6	401.0
P06	3.8	96.5	27.4	461.8

* EFFECTIVE MOMENT WEIGHT (ACTUAL MOMENT WEIGHT MINUS MOMENT WEIGHT OF SCREWS P07)

S-M56-SM-12978-00-B

Figure 7: Balance Screws

2.12.2 One-Shot Plot

(1) Make a record of the type and location of the original balance screws installed (see Figure 8).

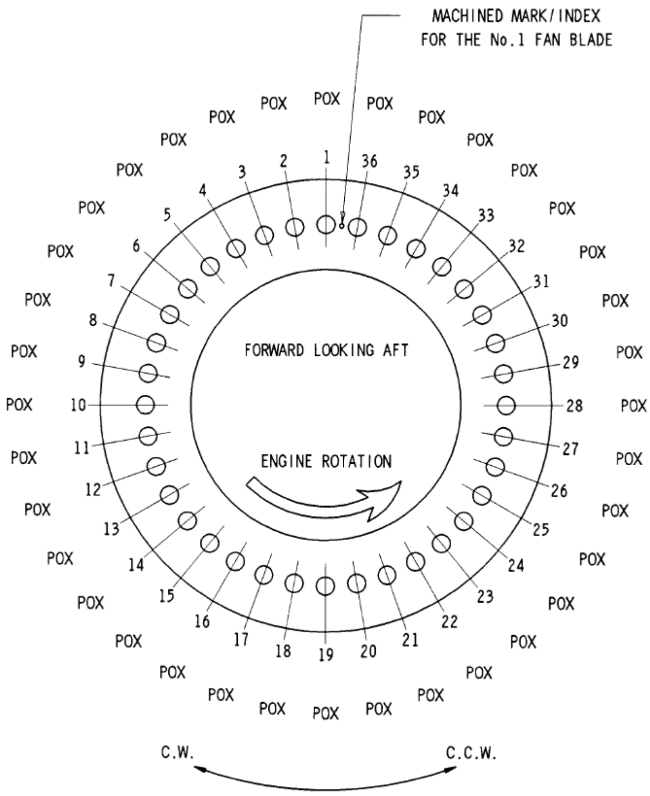


Figure 8. Balance Screw Location Chart

(2) Start the engine. Let it become stable at minimum idle for 5 minutes.

(3) Increase the speed to 80% N1 and let it become stable for 7 minutes to warm up the engine.

Caution: do not exceed 8 mils d.a. to avoid engine deterioration.

(4) Increase continuously the speed up to 94% N1. Decrease continuously the speed to the next lower N1 speed as given in Figure 9 while you monitor the No. 1 bearing and FFCC vibration indication. If vibration level is more than 8 mils D.A., decrease

N1 until the vibrations are 8 mils D.A., or less and do a prebalancing operation with speeds available with a vibration level of less than 8 mils D.A.

(5) Let it become stable at the points you had in above paragraph. Make sure that the vibrations become stable. To do this, monitor alternatively the No. 1 bearing and FFCC vibration indications. When it becomes stable, make a record of the vibration amplitudes (U0) and phase angles (A0) for each sensor and each speed. Use the one shot initial run recording worksheet. Refer to Figure 9.

CFM56-5C TRIM BALANCE
ANALYSER METHOD - RECORDING WORKSHEET

VIB. SENSOR	No.1 SPEED %	No.1 SPEED RPM	VIBRATION AMPLITUDE (MILS D.A) (U)o	SENSITIVITY (MULTIPLY BY)	BALANCE WEIGHT (CM.G) (W)	VIBRATION ANGLE (DEGREES) A0	PHASE LAG (DEGREES)	BALANCE WEIGHT ANGLE (DEGREES) A(W)	INDEX
No.1 BEARING	94	4500		350			240		
	90	4300		450			230		
	85	4000		350			220		
	73	3500		500			240		
	59	2800		450			60		
FFCC SENSOR	94	4500		350			260		
	90	4300		500			280		
	84	4000		600			280		
	73	3500		400			320		
	59	2800		650			40		

Figure 9: One-Shot Initial Run Recording Worksheet

(6) Decrease the power to idle, let it become stable for 5 minutes, then shut down.

(7) To calculate more easily, make a selection of the 3 highest indicated amplitudes for No. 1 bearing and for FFCC, thru the selection of 3 to 6 balance speeds.

You can automatically calculate all the data that you have written.

(8) Refer to worksheet Figure 9 and find the balancing weights and angles necessary for the 3 to 6 different speeds set before as follows:

(a) Balance weight is the result of amplitude U0 multiplied by sensitivity:

W = U0 x SENSITIVITY

(b) Calculated angle A(W) of balance weight is the result of vibration angle A0 plus the phase lag:

A = A0 + Phase lag

(9) Index the 6 selected points set from A to F.

(10) Give the applicable scale in g.cm to the graduations on a polargraph, Figure 10, so that you can plot the selection of points.

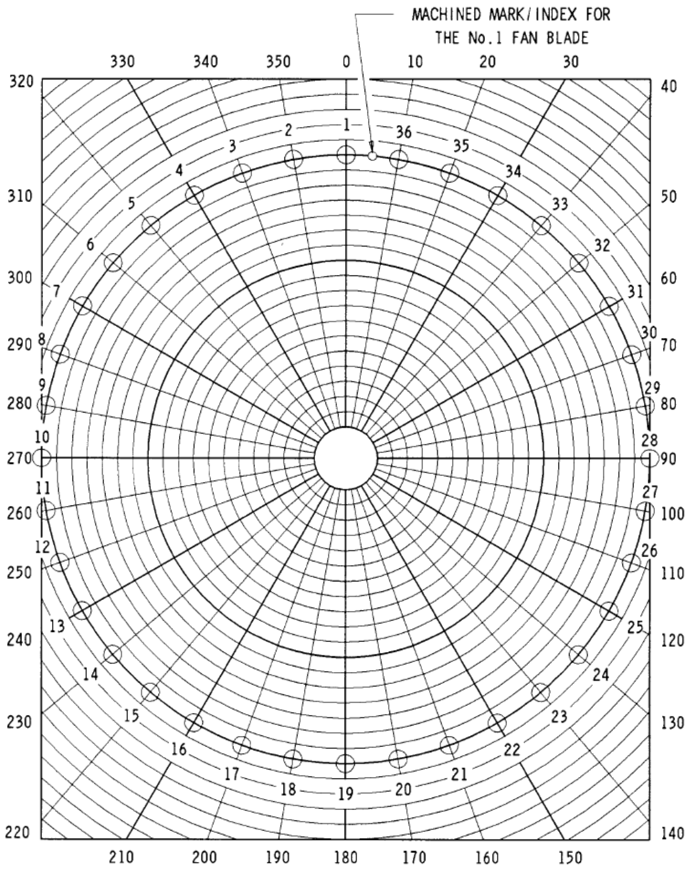


Figure 10: Polar Graph for Trim Balance

(11) Plot the 6 points indexed before (the moment weights W at their correct angle A(W), which is A0 + Lag angle).

(12) Take the length (measured in g.cm) between each pair of points indexed P1 and P2 in the Figure 11 and make a record with the one shot initial run optimization worksheet in vector length column of Figure 11.

VECTOR INDEX P1 P2	P1 P2 VECTOR LENGTH (G.LM)	VECTOR SENSITIVITY			AMPLITUDE RESIDUAL VIBRATION (P1 P2)/SP1+SP2 (MILS D.A.)	DISTANCE FROM P1 dP1=(P1 P2)xSP1 / (SP1+SP2)
		SENSIT. P1	SENSIT. P2	SP1+SP2		
A B						
A C						
A D						
A E						
A F						
B C						
B D						
B E						
B F						
C D						
C E						
C F						
D E						
D F						
E F						

Figure 11: One-Shot Initial Run Optimization Worksheet

(13) Write the related sensitivity of each vector end (P1 and P2) on worksheet, Figure 11, with data from worksheet, Figure 9, and add the 2 sensitivities for each vector.

(14) Calculate the minimum residual vibration amplitude of each pair of points:

Residual Vibration = The length of vector P1-P2 divided by the sum of the sensitivity of P1 plus the sensitivity of P2.

Formula: RV = P1/P2 divided by (SP1 + SP2)

(15) Make a selection of the pair of points with the maximum vibration amplitude and calculate the distance of the balance point to the vector first end (point P1).

The distance from P1 = P1/P2 X SP1 divided by (SP1 + SP2)

(16) Draw a line between selected points P1 and P2 and plot the location W1.

(17) Read the initial corrective weight value of W1 and phase angle A (W1).

(18) Calculate the residual vibration levels as follows:

Once you have found the imbalance correction weight in step (16) above, its position with respect to the indexed points A to F allows to determine the residual vibrations at those points (each sensor at each speed) as indicated in the following steps.

(a) Assign the appropriate scale in g.cm to graduations of a polar graph, Figure 10.

(b) Plot the 6 previously indexed points (the balance weights W at their respective angle A(W).

(c) On the same polar graph, plot the imbalance correction weight W1 at its angle A(W1) determine in step (16).

(d) Determine residual vibrations as follows:

1- Measure length (g.cm) between each indexed point A to F and selected imbalance correction weight W1.

2- For each point A to F, divide the length by sensitivity of indexed point to obtain residual vibration amplitude.

Example:

Residual vibration for A = Length between A and W1

divided by the sensitivity of A

(19) Optimize the residual vibration levels as follows:

NOTE: Once the residual vibrations have been calculated, some residual vibrations can be too high, therefore the position of the imbalance correction weight W1 must be improved to make the residual vibration amplitudes acceptable.

(a) Around each of the affected points draw a circle whose radius is equal to the sensitivity at that point multiplied by the desired maximum vibration amplitude. Radius (g.cm) = Sensitivity (g.cm divided by mils) X vibration (mils)

(b) Find on the polar graph an intersection point common to all the above circles. Refer to Figure 12.

(20) Find the original balance weight (W0) which agrees with the original balance screws written in step A.(1). Use Figures such as, Figure 13, Figure 14, and Figure 15.

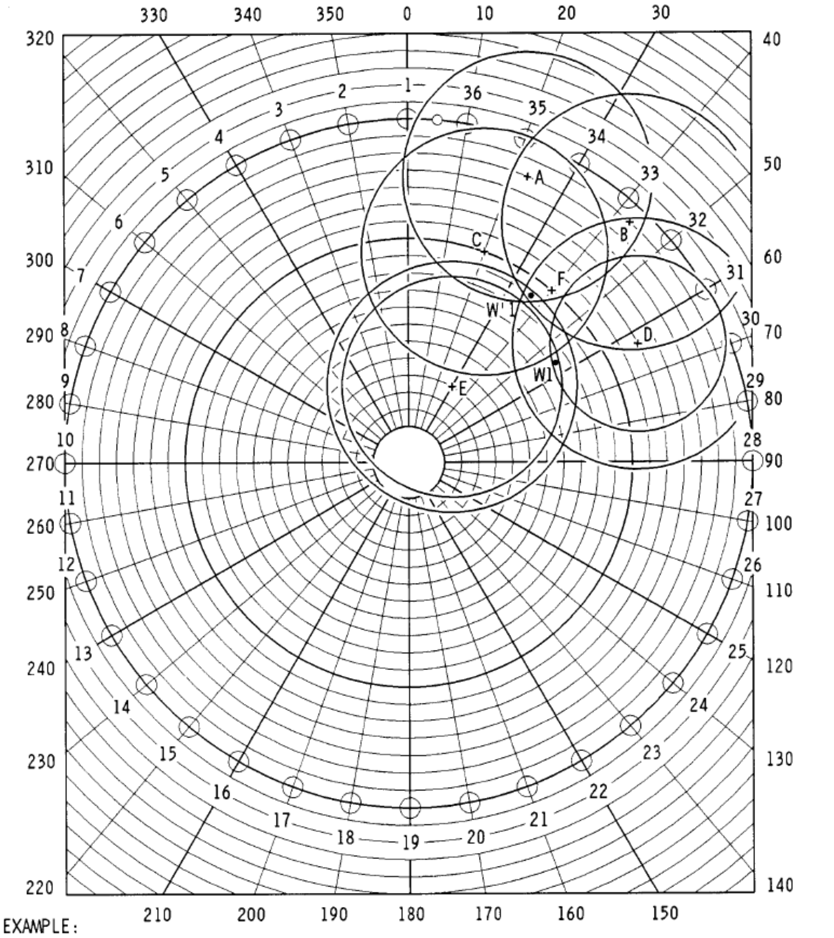
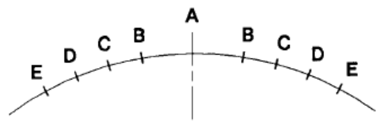
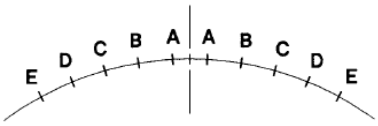


Figure 12: Optimization of Residual Vibration Levels



	A	B	C	D	E		A	B	C	D	E		A	B	C	D	E
1000	2	2	3	7	7	1250	6	5	7	7	7	2500	5	5	7	5	5
1010	3	7	5	7	7	1300	1	7	4	4	7	2550	6	4	4	6	7
1020	7	7	2	5	7	1350	2	4	7	7	4	2600	6	7	4	6	6
1030	5	7	4	7	7	1400	7	5	7	7	5	2650	6	6	3	6	7
1040	2	7	6	7	7	1450	7	7	6	4	7	2700	6	5	5	5	7
1050	7	2	7	7	6	1500	3	4	7	4	7	2750	5	5	6	5	7
1060	5	4	7	7	7	1550	7	7	5	6	7	2800	6	5	5	6	7
1070	5	2	2	7	7	1600	7	6	7	5	7	2850	7	5	5	5	5
1080	1	3	3	7	7	1650	3	6	3	7	7	2900	7	6	3	6	6
1090	4	7	5	7	7	1700	4	7	5	7	5	2950	4	5	4	4	5
1100	2	2	7	4	7	1750	4	4	5	7	7	3000	5	4	4	5	5
1110	1	7	6	1	7	1800	3	5	5	7	7	3050	7	5	5	6	6
1120	7	3	7	7	5	1850	6	6	3	7	7	3100	6	6	4	4	4
1130	1	1	6	7	7	1900	2	7	4	4	4	3150	4	4	6	4	6
1140	7	2	7	6	7	1950	2	6	6	7	7	3200	6	6	2	6	6
1150	4	4	7	1	7	2000	6	6	4	7	7	3250	5	5	5	5	5
1160	7	2	7	2	4	2050	6	5	7	6	7	3300	6	4	6	6	4
1170	7	7	6	2	7	2100	7	6	7	6	3	3350	5	5	5	6	5
1180	5	7	1	7	5	2150	7	7	5	5	6	3400	4	6	6	4	6
1190	5	5	7	7	7	2200	7	5	7	6	5	3450	2	6	6	6	6
1200	5	7	7	6	7	2250	7	5	5	7	6	3500					
1210	7	2	6	7	7	2300	6	6	3	3	7						
1220	7	2	3	7	3	2350	7	5	6	5	7						
1230	2	5	7	7	2	2400	5	5	4	4	7						
1240	5	7	7	3	3	2450	3	6	3	6	7						

Figure 13: Balance Weight Centered on one Screws



	A	B	C	D	E		A	B	C	D	E		A	B	C	D	E
1000	2	7	7	7	6	1250	7	5	7	7	4	2500	2	5	5	5	7
1010	3	3	7	7	7	1300	7	4	7	7	6	2550	4	4	7	6	4
1020	7	7	7	5	3	1350	3	7	6	7	7	2600	3	3	6	6	7
1030	7	7	4	3	7	1400	3	7	3	3	7	2650	5	4	5	7	4
1040	1	3	7	7	3	1450	5	7	7	7	6	2700	4	4	5	5	7
1050	3	1	3	7	7	1500	7	5	5	7	7	2750	5	7	5	5	5
1060	7	5	7	2	7	1550	2	2	2	4	7	2800	6	4	7	6	4
1070	1	6	7	7	7	1600	3	7	5	7	3	2850	5	5	5	4	7
1080	6	7	1	7	7	1650	3	5	7	7	3	2900	6	3	6	7	6
1090	6	1	7	7	7	1700	4	7	4	3	7	2950	5	5	5	5	7
1100	2	7	3	2	7	1750	3	5	3	7	7	3000	6	5	7	5	6
1110	7	7	2	4	2	1800	6	7	7	6	1	3050	5	5	5	6	7
1120	2	5	7	7	7	1850	1	4	7	4	4	3100	5	6	7	6	6
1130	7	7	7	5	4	1900	6	7	6	1	7	3150	4	6	6	6	7
1140	1	3	3	7	7	1950	7	7	5	5	5	3200	6	6	6	4	7
1150	7	3	7	7	6	2000	3	7	6	7	6	3250	3	3	6	6	6
1160	3	4	7	7	7	2050	4	5	4	7	7	3300	6	6	6	7	6
1170	3	7	7	5	7	2100	4	5	7	5	7	3350	4	4	6	4	6
1180	3	7	7	3	2	2150	4	4	6	7	7	3400	6	6	6	6	7
1190	6	7	7	1	1	2200	5	7	6	7	5	3450					
1200	7	4	7	4	7	2250	6	3	6	7	7	3500					
1210	7	3	3	7	2	2300	5	5	5	7	7						
1220	4	7	7	4	7	2350	6	5	7	5	7						
1230	7	7	7	6	4	2400	6	4	6	7	7						
1240	3	7	5	7	7	2450	6	5	7	6	7						

Figure 14: Balance Weight Centered Between Two Screws

NUMBER OF SCREWS	P01	P02	P03	P04	P05	P06
1 SCREW	89.3	174.3	257.5	336.2	401	461.8
2 SCREWS	177.9	347.2	512.9	669.8	798.9	920.1
3 SCREWS	265.2	517.5	764.6	998.3	1191	1371
4 SCREWS	350.4	683.9	1010	1319	1573	1812
5 SCREWS	433	845.1	1248	1630	1944	2239
6 SCREWS	512.3	999.8	1477	1929	2300	2649
7 SCREWS	587.7	1147	1694	2212	2639	3039
8 SCREWS	658.6	1285	1899	2479	2957	3406
9 SCREWS	724.6	1414	2089	2727	3253	
10 SCREWS	784.9	1532	2263	2955	3524	
11 SCREWS	839.3	1638	2420	3160		
12 SCREWS	887.3	1732	2558	3340		
13 SCREWS	928.6	1812	2677	3496		
14 SCREWS	962.8	1879	2775			
15 SCREWS	989.7	1931	2853			
16 SCREWS	1009	1969	2908			
17 SCREWS	1021	1992	2943			
18 SCREWS	1025	2000	2953			

Figure 15: Correspondene Between Screws and Balance Weights

(21) Find the related phase angle A(W0) with Figure 16.

CALCULATED PHASE ANGLE (A)	LOCATION TO PLACE CORRECTION WEIGHTS	CALCULATED PHASE ANGLE (A)	LOCATION TO PLACE CORRECTION WEIGHTS
3-7	36-1	93-97	27-28
8-12	36	98-102	27
13-17	35-36	103-107	26-27
18-22	35	108-112	26
23-27	34-35	113-117	25-26
28-32	34	118-122	25
33-37	33-34	123-127	24-25
38-42	33	128-132	24
43-47	32-33	133-137	23-24
48-52	32	138-142	23
53-57	31-32	143-147	22-23
58-62	31	148-152	22
63-67	30-31	153-157	21-22
68-72	30	158-162	21
73-77	29-30	163-167	20-21
78-82	29	168-172	20
83-87	28-29	173-177	19-20
88-92	28	178-182	19

S-M56-SM-01437-01-B

Figure 16: Location to Install Correction Weights

(22) Plot vector Wo at angle A(W0) on a polar graph with graduations

(23) Starting from the end of W0, plot vector W'1 at angle A(W'1), found in step (19).

(24) Add the 2 vectors. To do this, make a line which starts from the origin of graph to the end of vector W'1.

(25) Read the related imbalance value WS and phase angle AS.

(26) Find if the weight location is centered on the screw hole or between holes (refer to Figure 16).

(27) Make a selection of the part numbers of balance screws from Figures such as, Figure 13, Figure 14, and Figure 15. Write the balance screws with their location on a balance screw location chart (Figure 8).

(28) Remove the original balance screws and install the new balance screws found in step above. Torque the screws to 70-75 lb in. (8-8,5 N.m.)

(29) Start the engine and let it become stable at idle for 5 minutes.

2.12.3 Vector Analysis

The vector analysis is used when the "one shot" procedure does not operate because the phase lag and/ or sensitivity are not accurate. The vector analysis gives accurate phase lag and sensitivity that agree with the engine/analyzer system. If you calculate again with corrected phase lag and sensitivity it will generally give a correct balancing.

To calculate more easily, a minimum of 3 vibration data for No. 1 bearing and 3 vibration data for FFCC is strongly recommended. If an automatic calculation is used, calculate with all the vibration data.

(1) Use the polar graph to get a first scale from the center of the graph to the outer perimeter in mils D.A. and a second one in g.cm.

(2) After selection, for each of the highest vibrations, do as follows. Refer to Figure 17 .

(a) Plot the initial run vector (U0) at angle AO. Refer to Figure 9.

(b) Plot the first test run vector U1, that corresponds to the same speed and sensor as vector U0, at angle A1, as written. Refer to Figure 17.

NOTE: Vectors U0 and U1 are in mils D.A.

(c) Plot the initial balance weight W'1 at its correct angle location A(W'1). W'1 is in g.cm.

(d) Make a vector B from the tip of vectors U0 to the tip of vector U1.

(e) Make a vector B', equal in length and direction to vector B, starting at origin of graph.

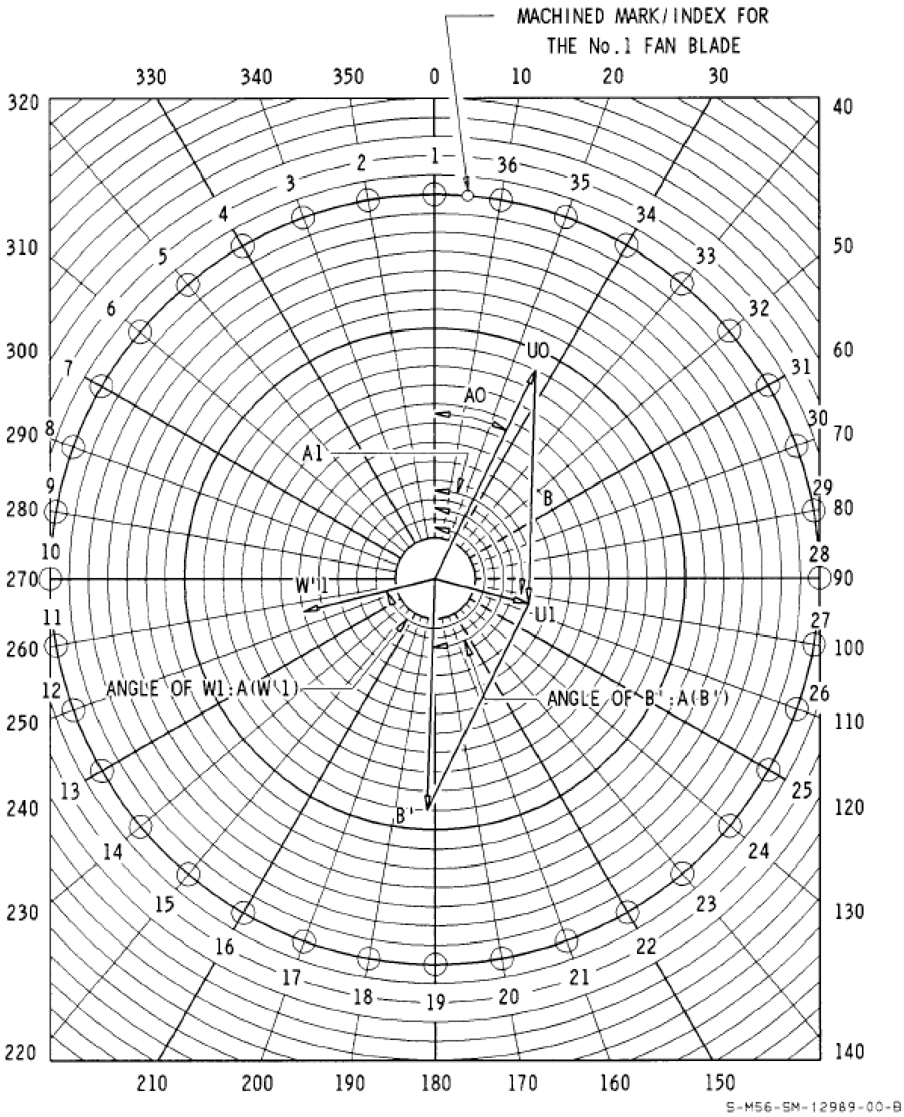


Figure 17: Vectorial polagraph

(3) Find the phase lag of each sensor at each speed as follows: Phase lag = (Angle of vector W'1) minus (Angle of vector B') plus 180° Phase lag = A(W'1) - A(B') = 180°

(4) Find the sensitivity of each sensor for each speed as follows: Sensitivity = W'1 (g.cm) divided by vector B (mils)

(5) Fill in a recording worksheet, Figure 18, with the calculated phase lag and sensitivity for each sensor at each speed.

CFM56-5C TRIM BALANCE
ANALYSER METHOD - RECORDING WORKSHEET

VIB. SENSOR	N1 SPEED %	N1 SPEED RPM	VIBRATION AMPLITUDE (MILS D.A) (U1)	SENSITI-VITY (MUL-TIPLY BY)	BALANCE WEIGHT (CM.G) (W)	VIBRATION ANGLE (DEGREES) (A1)	PHASE LAG (DEGREES)	BALANCE WEIGHT ANGLE (DEGREES) A(W)	INDEX
No.1 BEARING	94	4500							
	90	4300							
	84	4000							
	73	3500							
	59	2800							
CCFF SENSOR	94	4500							
	90	4300							
	84	4000							
	73	3500							
	59	2800							

Figure 18: Vector Analysis Recording Worksheet

(6) Transfer the vibration values written during the one-shot initial run, to the recording worksheet which has the accurate phase lag and sensitivity found in step (4) above.

(7) Refer to worksheet and find the balancing weights and angles necessary for the 3 to 6 different speeds set before as follows:

(a) Balance is the result of amplitude U0 multiplied by accurate sensitivity:

W = U0 x accurate sensitivity

(b) Calculated angle A(W) of balance weight is the result of vibration angle A0 + accurate phase lag.

(8) Index the 6 selected points set from A to F.

(9) Give the applicable scale in g.cm to the graduations on a polargraph so that you can plot the selection of points.

(10) Plot the 6 points indexed before (the moment weights W at their correct angle A(W) which is A0 + Lag angle).

(11) Take the length (measured in g.cm) between each pair of points indexed P1 and P2 in a worksheet like Figure 11 , and make a record with the vectorial analysis optimization worksheet in vector length column.

(12) Write the related accurate sensitivity (calculated in step (4) above), of each vector ends (P1 and P2) on worksheet, and add the 2 sensitivities for each vector.

(13) Calculate the minimum residual vibration amplitude of each pair of points:

Residual Vibration = The length of vector P1-P2 divided by the sum of the sensitivity of P1 plus the sensitivity of P2.

(14) Make a selection of the pair of points with the maximum vibration amplitude and calculate the distance (in g.cm) of the balance point to the vector first end (point P1):

(15) Draw a line between points P1 and P2 and plot the calculated length dP1. Start at point P1 of selected vector P1 P2 and find the vectorial analysis balance weight location W2.

(16) Read the imbalance correction weight value of W2 and angle A(W2).

(17) Calculate the residual vibration levels as follows:

NOTE: Once you have found the imbalance correction weight in step (16) above, its position with respect to the indexed points A to F allows to determine the residual vibrations at those points (each sensor at each speed) as indicated in the following steps.

(a) Assign the appropriate scale in g.cm to graduations of a polar graph.

(b) Plot the 6 previously indexed points (the balance weights W at their respective angle A(W) determined in step (7) above.

(c) On the same polar graph, plot the imbalance correction weight W2 at its angle A(W2) determined in step (16).

(d) Determine residual vibrations as follows:

1- Measure length (g.cm) between each indexed point A to F and selected imbalance correction weight W2.

2- For each point A to F, divide the length by sensitivity of indexed point to obtain residual vibration amplitude.

(18) Optimize the residual vibration levels as follows:

NOTE: Once the residual vibrations have been calculated, some residual vibrations can be too high, therefore the position of the imbalance correction weight W must be improved to make the residual vibration amplitudes acceptable.

(a) Around each of the affected points draw a circle whose radius is equal to the sensitivity at that point multiplied by the desired maximum vibration amplitude. Radius (g.cm) = Sensitivity (g.cm) divided by mils X Vibration (mils)

(b) Find on the polar graph an intersection point common to all the above circles.

(19) Find the original balance weight W0 which agrees with the original balance screws written in step (1) of one-shot precedure, determined in step (19) of one-shot precedure.

(20) Find the related phase angle A(W0) determined in step (20) of one-shot precedure.

(21) Plot vector W0 at angle A(W0) on a polar graph.

(22) Starting from the end of W0, plot vector W'2 at angle A(W'2) found in step (18) above.

(23) Add the 2 vectors. To do this, make a line which starts from the origin of graph to the end of vector W'2.

(24) Read the related imbalance value WS and phase angle AS.

(25) Find if the weight location is centered on screw hole or between holes.

(26) Make a selection of the part numbers of balance screws from related figures. Write the balance screws with their location on a balance screw location chart.

(27) Remove the original balance screws and install the new balance screws found in step above. Torque the screws to 70-75 lb in. (8-8,5 N.m).

(28) Start the engine and let it become stable at idle for 5 minutes.

(29) Increase the power to 80% N1 and let it become stable for 7 minutes to warm up the engine.

(30) Increase continuously the speed up to 94% N1 while you monitor the No. 1 bearing and FFCC vibration indication. When it becomes stable, make a record of the vibration amplitude (U2) and phase angles (A2) for each sensor and each speed. Use a recording worksheet.

(31) Decrease the power to idle, let it become stable for 5 minutes then shut down.

(32) Make sure that the No. 1 bearing and the FFCC vibration level is within 4 Mils D.A. If the vibration amplitudes are within limits, the trim balance procedure is now complete. If the imbalances are still more than the necessary levels, do an intervectorial analysis as mentioned in fan trim balance engine manu

APPENDIX “A”

DRAWINGS OF TEST BENCH SECTIONS AND PLANS

Место размещения стенда. Схема генерального плана. М1:500
Bench location. Site Plan Layout. Scale 1:500

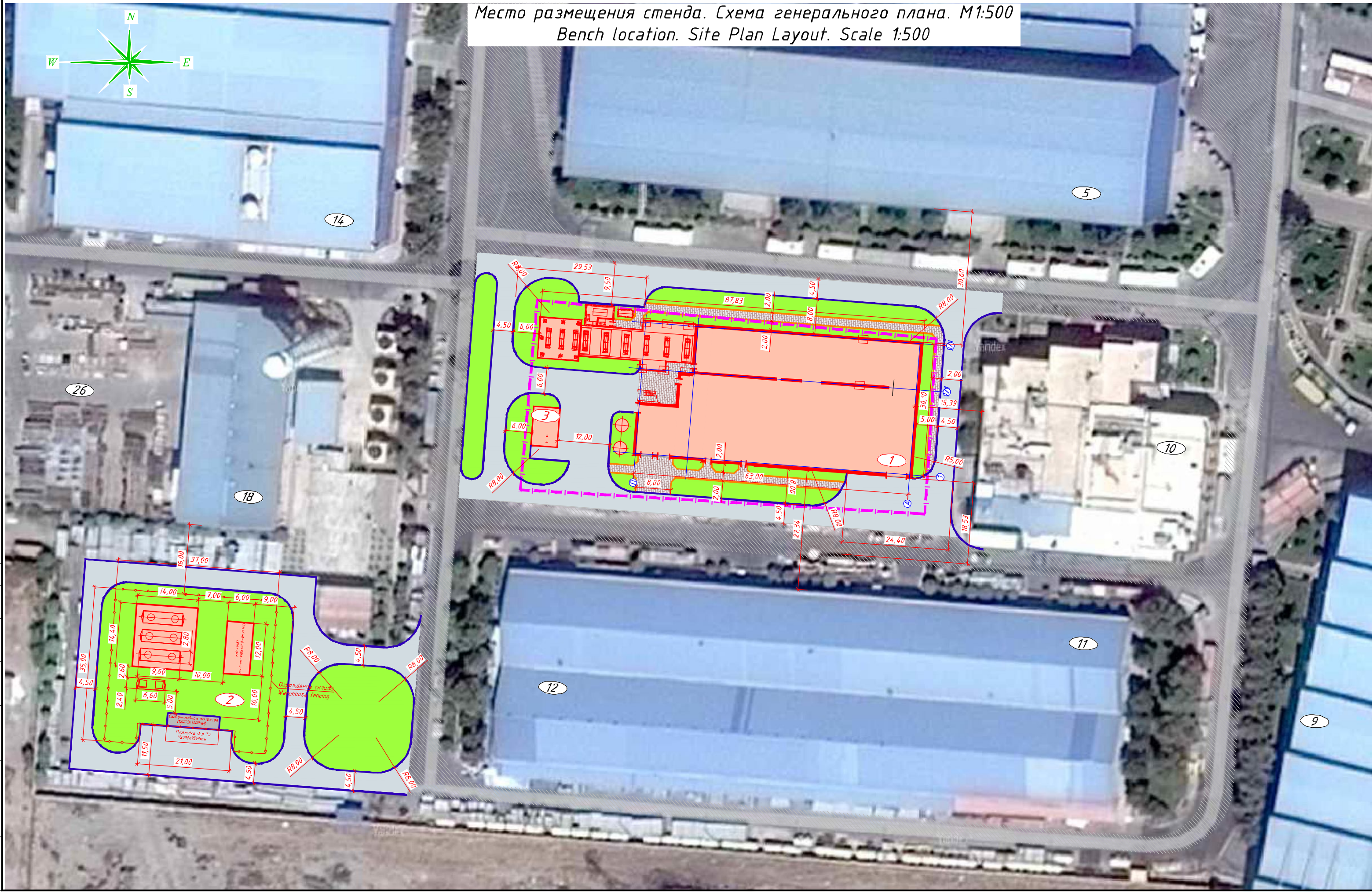
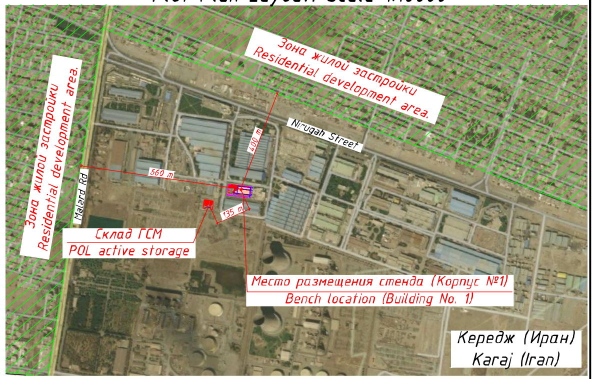


Схема ситуационного плана. М1:10000
Plot Plan Layout. Scale 1:10000



Экспликация зданий и сооружений
Explanation of buildings and structures

Номер по генплану Item No. in the Site Plan.	Наименование Denomination	Примечание Remarks
1	Корпус №1 Building No. 1	Проектируемый At the design stage
2	Расходный склад ГСМ корпуса №1 POL active storage in building No. 1	Проектируемый At the design stage
3	Контейнерная трансформаторная подстанция (КТП (РУ)) Complete transformer substation (CTS)	Проектируемая At the design stage
5	Корпус газовых турбин Gas turbines building	Существующий Existing
9	Корпус паровых турбин Steam turbines building	Существующий Existing
10	Ресторан (столовая) Dining establishment (canteen)	Существующий Existing
11, 12	Спортивный зал Gym	Существующий Existing
14	Стенд испытания газовой турбины Gas turbine test bench	Существующий Existing
18, 26	Склад химических материалов Chemicals warehouse	Существующий Existing

Условные обозначения:
Legend:

- граница землепользования, площадь – 3862,5 м^2
- land use boundaries – area – 3862.5 m^2

- проектируемая застройка
- designed development area

- проектируемые дороги, проезды и площадки с твердым покрытием (с бортовым камнем)
- designed paved roads, passages and sites (with border stone)

- проектируемые тротуары, пешеходные дорожки и площадки (с бортовым камнем)
- designed walkways, pedestrian ways and sites (with border stone)

- проектируемое озеленение
- designed landscaping

Место размещения стенда
Test bench location.
Раздел 5 Концептуального проекта
Section 5. Conceptual design.

ГИПРОНИИ АВИАПРОМ

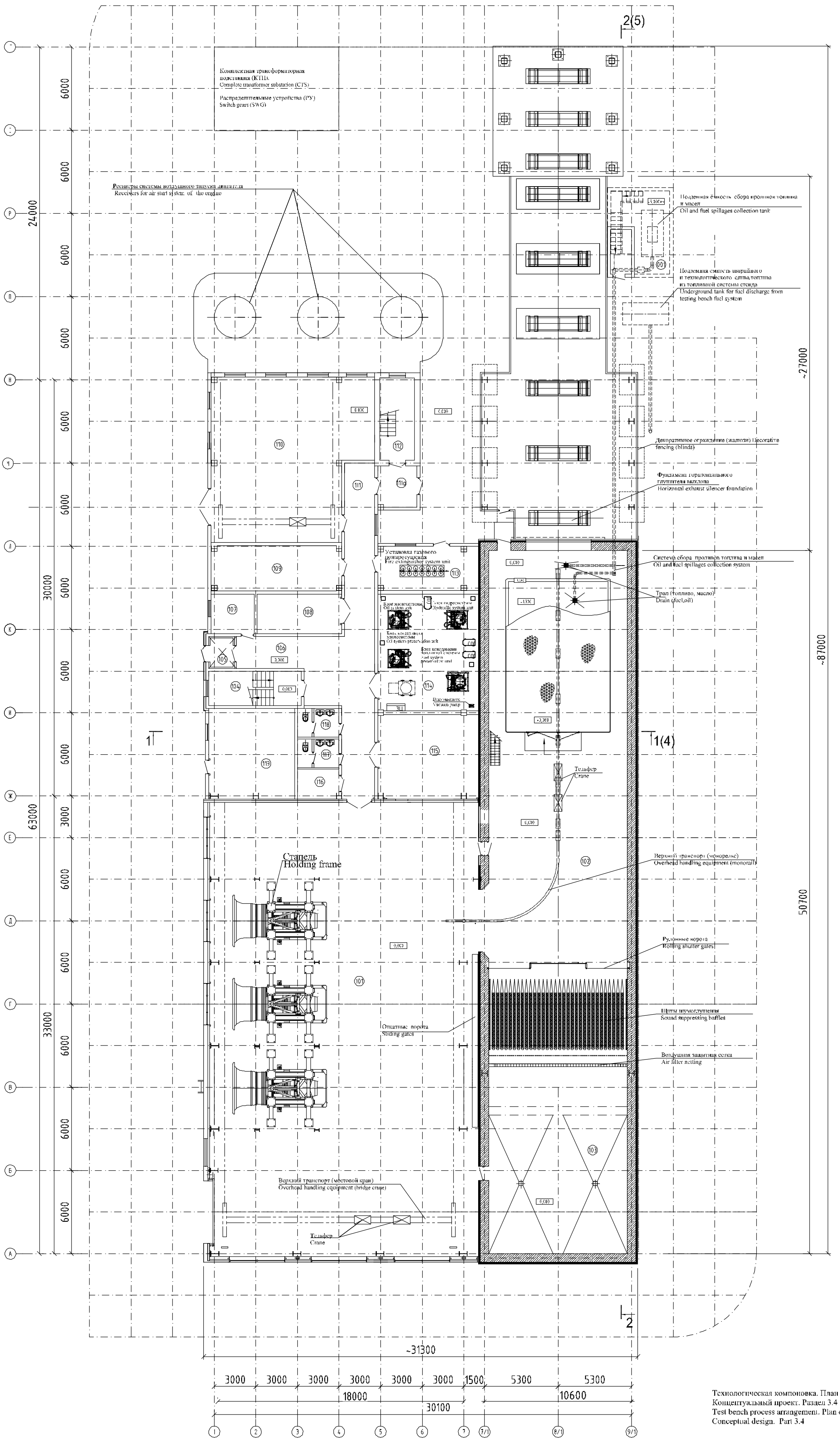
Согласовано

Инв.№подл. | Подпись и дата | Взам.инв.№

Заказ № (Order No.) 364-Д001 Формат (Format) 3А3

Испытательная станция авиационного двигателя.
Aircraft engine test bench.

Технологическая компоновка. План 1-го этажа.
Test bench process arrangement. Plan of ground floor at elevation of 0,000 ("1-st floor")
M1:100



Экспликация помещений 1-го этажа на отм. 0,000				
Explanation of rooms - ground floor at elevation of 0,000 ("1-st floor")				
№ помещения	Наименование	Площадь помещения, кв.м	Категория помещения по СП12.13130	Категория помещения по ПУЭ
Room №	Denomination	Room area, sq.m.	Fire hazard room category	EIR room category
101	Зал подготовки	633,7	В3	П-I
	Preparation area			
102	Испытательный бокс двигателя	507,0	В2	П-I
	Testing chamber			
103	Шахта всасывания		Д	-
	Inlet			
104	Лестничная клетка	16,7		
	Stairway			
105	Тамбур	3,6		
	Frontdoor antechamber			
106	Коридор	25,2		
	Corridor			
107	Телекоммуникационная	8,9		
	Telecommunication room			
108	Кладовая инструмента	17,3	В4	П-IIа
	Equipment storage room			
109	Электрощитовая	28,8	В4	П-IIа
	Switchboard room			
110	Компрессорная системы воздушного запуска двигателя	124,2	В3	П-I
	Air start system compressor room			
111	Коридор	50,6		
	Corridor			
111а	Тамбур	17,4		
	Backdoor antechamber			
112	Лестничная клетка	14,8		
	Stairway			
113	Помещение газового пожаротушения	23,1	Д	-
	Fire extinguisher system room			
114	Маслокомната	60,9	В1	П-I
	Oil room			
115	Электротехническое помещение	44,0	В4	П-IIа
	Electric units room			
116	Кладовая уборочного инвентаря	6,4		
	Cleaning facilities storage room			
117	Санузел	6,2		
	Water closet			
118	Санузел	6,2		
	Water closet			
119	Тепловой пункт	40,3		
	Heat supply station			

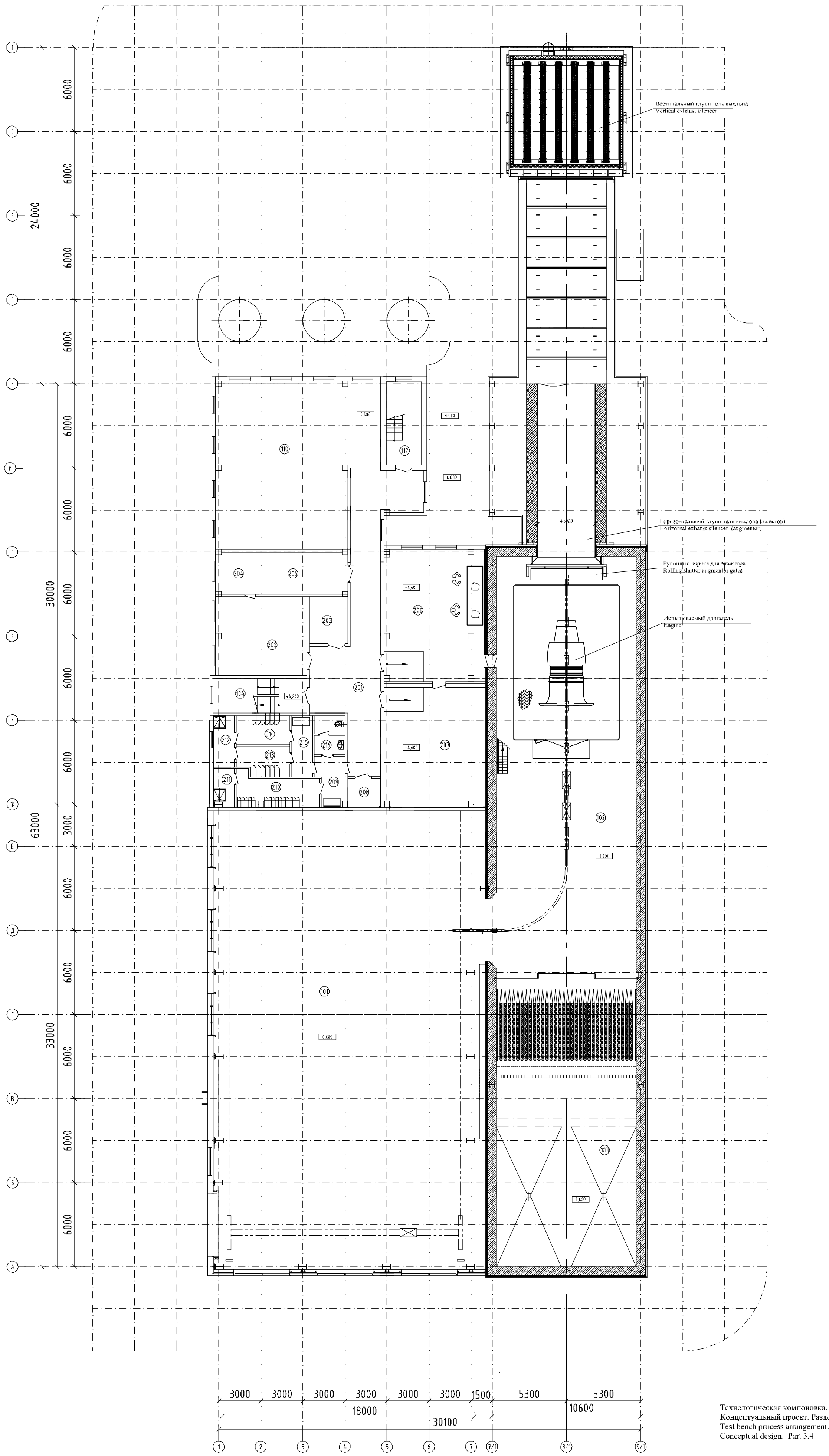
Экспликация помещений подземной камеры для ёмкости сбора проливов топлива и масел (с пола)				
Explanation of rooms - basement for oil and fuel spillages collection tank room				
№ помещения	Наименование	Площадь помещения, кв.м	Категория помещения по СП12.13130	Категория помещения по ПУЭ
Room №	Denomination	Room area, sq.m.	Fire hazard room category	EIR room category
001	Помещение ёмкости сбора проливов топлива и масел	25,5	А	В-Iа
	Oil and fuel spillages collection tank room			

Технологическая компоновка. План 1-го этажа.
Концептуальный проект. Раздел 3.4
Test bench process arrangement. Plan of ground floor at elevation of 0,000 ("1-st floor")
Conceptual design. Part 3.4

ГИПРОНИИ АВИАПРОМ

Испытательная станция авиационного двигателя.
Aircraft engine test bench.

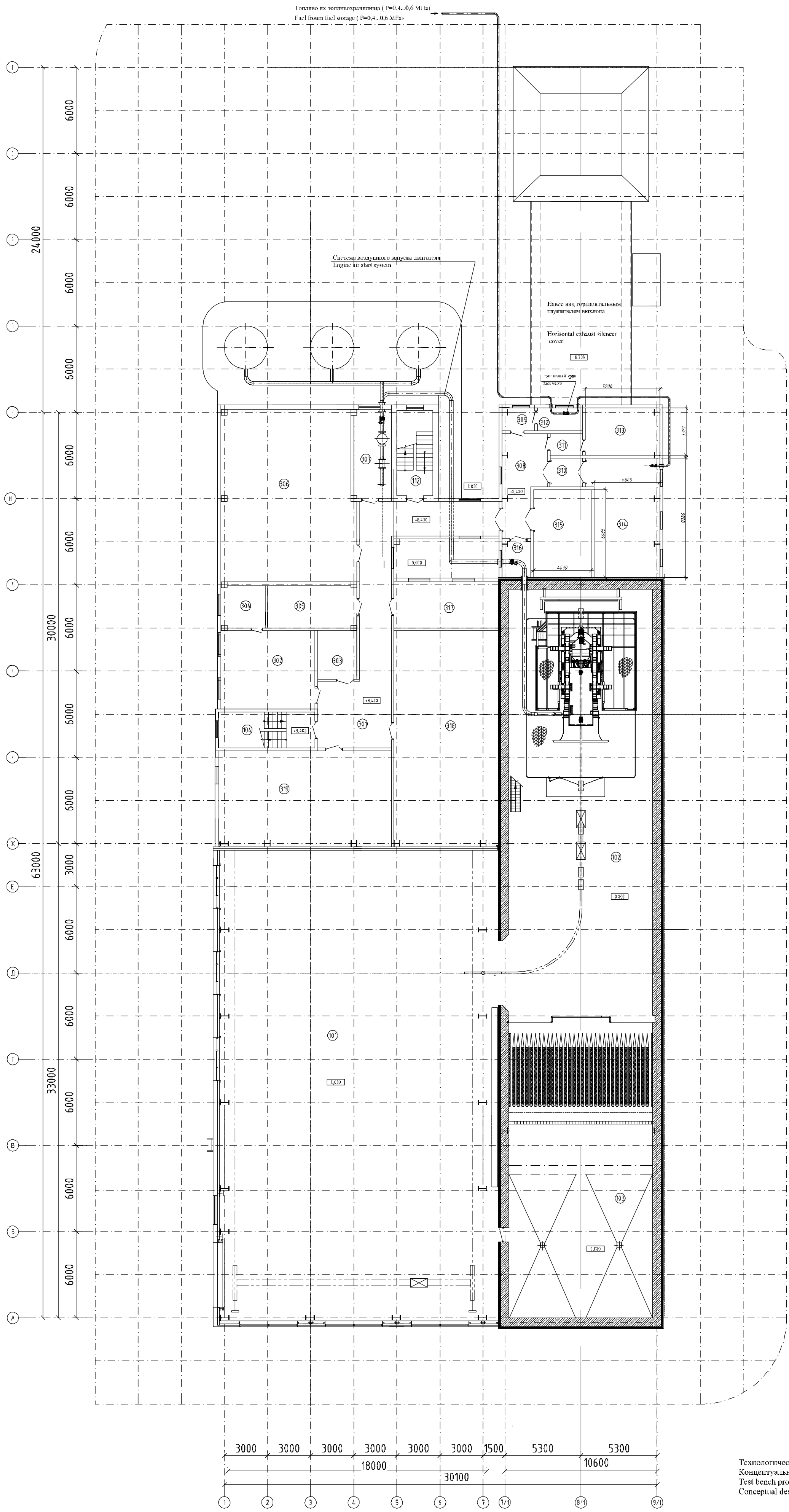
Технологическая компоновка. План 2-го этажа.
Test bench process arrangement. Plan of 1-st floor at elevation of +4,200 ("2-nd floor")
М1:100



Экспликация помещений 2-го этажа на отм. +4,200				
Explanation of rooms - 1-st floor at elevation of +4,200 ("2-nd floor")				
№ помещения	Наименование	Площадь помещения, кв.м	Категория помещения по СП12.13130	Категория помещения по ПУЭ
Room №	Denomination	Room area, sq.m.	Fire hazard room category	EIR room category
101	Зал подготовки		В3	П-I
	Preparation area			
102	Испытательный бокс двигателя		В2	П-I
	Testing chamber			
103	Шахта всасывания		Д	-
	Inlet			
104	Лестничная клетка	16,7		
	Stairway			
110	Компрессорная системы воздушного запуска двигателя		В3	П-I
	Air start system compressor room			
112	Лестничная клетка	14,8		
	Stairway			
201	Коридор	71.9		
	Corridor			
202	Зал обработки результатов испытаний	36,4		
	Test data processing room			
203	Архив	9,0	В3	П-IIа
	Archives			
204	Телекоммуникационная	9,0		
	Telecommunication room			
205	Электрощитовая	18,1		
	Switchboard room			
206	Кабина управления	68,3	В4	П-IIа
	Control room			
207	Помещение АСУТП	62,0	В4	П-IIа
	APCS room			
208	Комната уборочного инвентаря	4,8		
	Cleaning equipment storage room			
209	Тамбур	6,9		
	Antechamber			
210	Гардеробная мужская"1а","1б"	12,9		
	Men lockers "1а", "1б"			
211	Душевая	4,1		
	Shower room			
212	Душевая	5,5		
	Shower room			
213	Гардеробная мужская "1в" (6 шкафов для чистой одежды)	7,5		
	Гардеробная мужская "1в" (6 clean clothes lockers)			
214	Гардеробная мужская "1в" (6 шкафов для рабочей одежды)	8,0		
	Men lockers "1в" (6 work clothes lockers)			
215	Коридор	6,5		
	Corridor			
216	Санузел	6,0		
	Water closet			

Испытательная станция авиационного двигателя.
Aircraft engine test bench.

Технологическая компоновка. План 3-го этажа.
Test bench process arrangement. Plan of 2-nd floor at elevation of +8,400 ("3-rd floor")
M1:100



№ помещения	Наименование	Площадь помещения, кв.м	Категория помещения по СП12.13130	Категория помещения по ПУЭ
Room №	Denomination	Room area, sq.m.	Fire hazard room category	EIR room category
101	Зал подготовки		В3	П-I
	Preparation area			
102	Испытательный бокс двигателя		В2	П-I
	Testing chamber			
103	Шахта всасывания		Д	-
	Inlet			
104	Лестничная клетка	16,7		
	Stairway			
112	Лестничная клетка	14,8		
	Stairway			
301	Коридор	83,6		
	Corridor			
302	Технологическое бюро	35,5		
	Production engineering office			
303	Кладовая технологического оборудования	9,2	В4	П-IIа
	Process equipment storage room			
304	Телекоммуникационная	9,0		
	Telecommunication room			
305	Электрощитовая	17,0	В4	П-IIа
	Switchboard room			
306	Помещения систем охлаждения	110,6	В4	П-IIа
	Cooling system rooms			
307	Помещение воздушной системы запуска двигателя	16,7	В4	П-IIа
	Air start system process room			
308	Коридор	18,2		
	Corridor			
309	Тамбур помещения топливного крана	3,6		
	Fuel valve room airlock			
310	Тамбур помещения топливной системы	4,5		
	Fuel system room airlock			
311	Тамбур венткамеры топливной системы	3,4		
	Fuel system vent. chamber airlock			
312	Помещение топливного крана		А	В-Iа
	Fuel valve room			
313	Венткамера топливной системы	17,2	А	В-Iа
	Fuel system ventilation chamber			
314	Помещение топливной системы	38,9	А	В-Iа
	Testing bench fuel system process room			
315	Помещение системы загрузки гидронасосов	24,3	В2	П-I
	Hydraulic pump loading system room			
316	Помещение клапана воздушного запуска	5,3	В3	П-I
	Engine air start system valve room			
317	Помещение загрузки генераторов	23,5	В4	П-IIа
	Generators loading system process room			
318	Вытяжная венткамера	108,6		
	Exhaust ventilation camera			
319	Приточная венткамера	80,0		
	Supply ventilation camera			

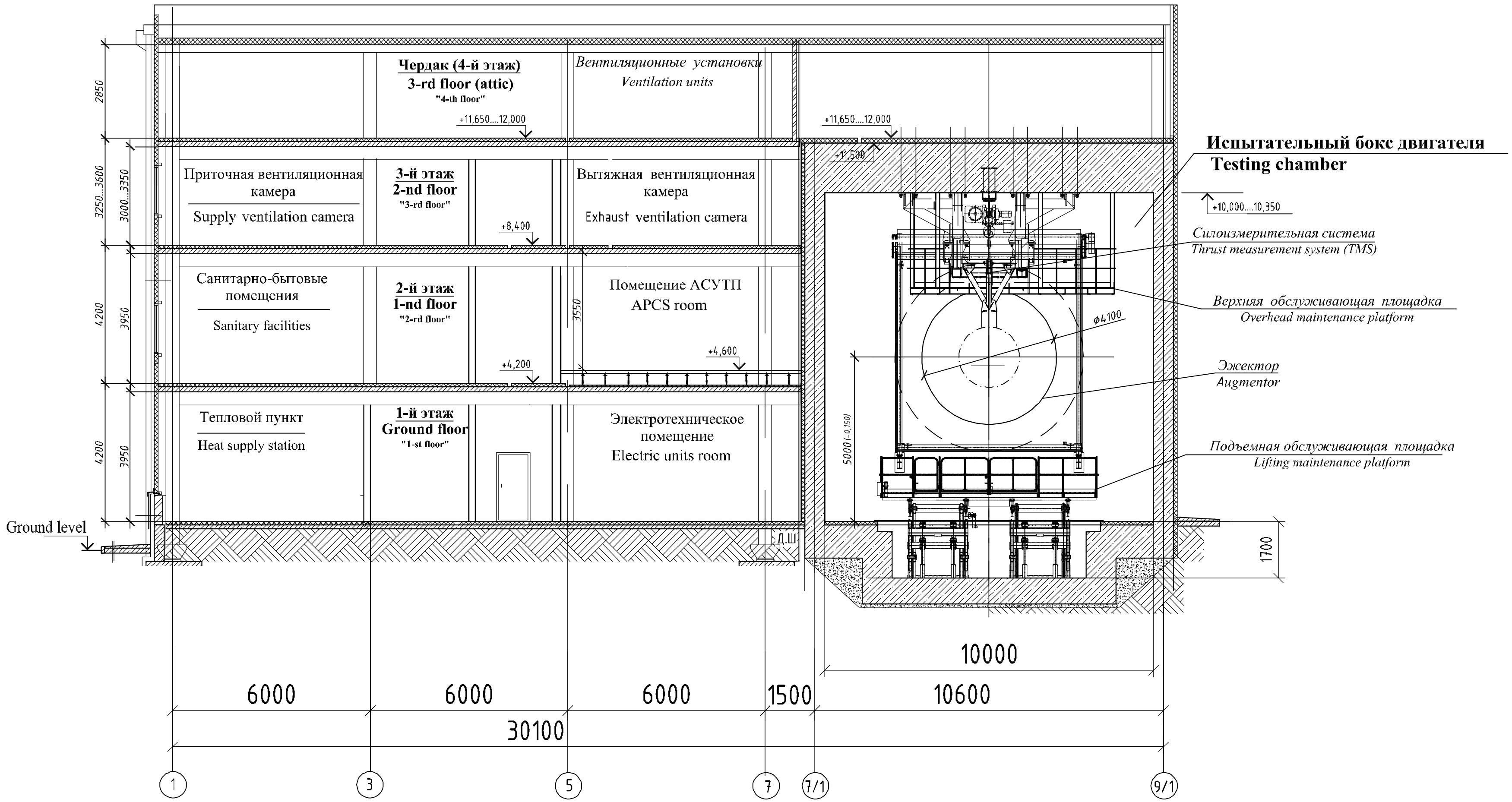
Экспликация помещений 3-го этажа на отм. +8,400
Explanation of rooms - 3-rd floor at elevation of 8,400 ("3-rd floor")

Технологическая компоновка. План 3-го этажа.
Концептуальный проект. Раздел 3.4
Test bench process arrangement. Plan of 2-st floor at elevation of 8,400 ("3-rd floor")
Conceptual design. Part 3.4

ГИПРОНИИ АВИАПРОМ

Испытательная станция авиационного двигателя.
Aircraft engine test bench.

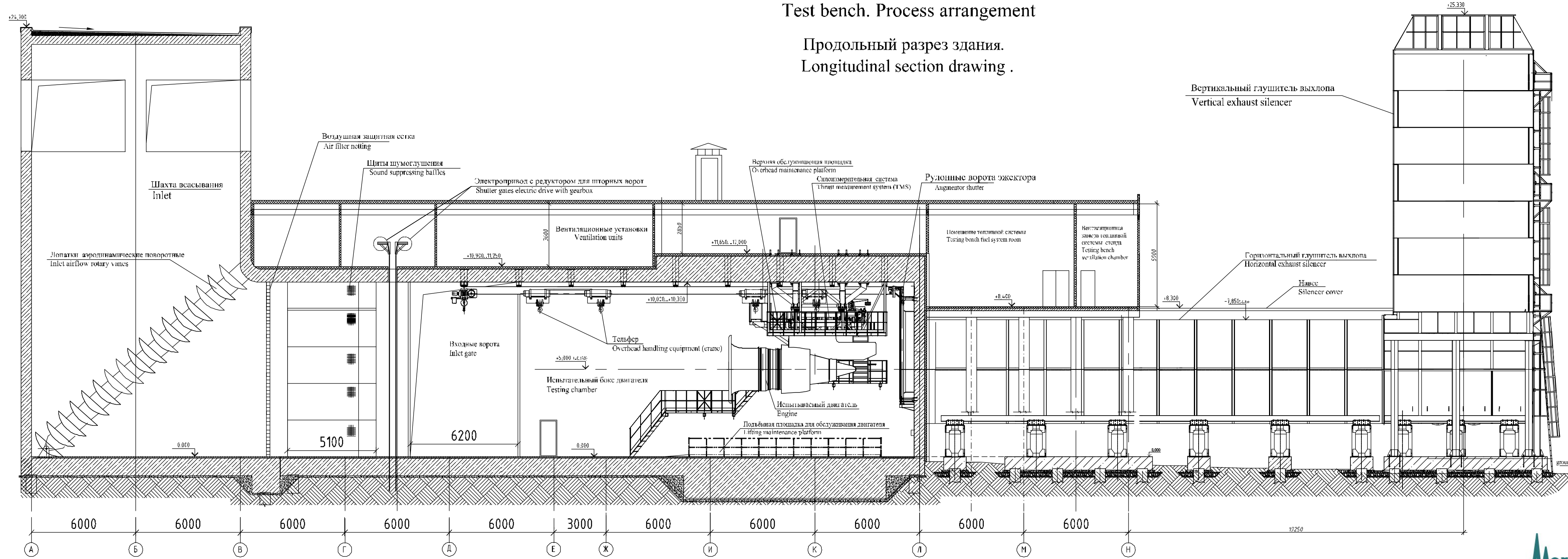
Технологическая компоновка. Поперечный разрез здания.
Test bench process arrangement. Cross-section drawing
M1:100



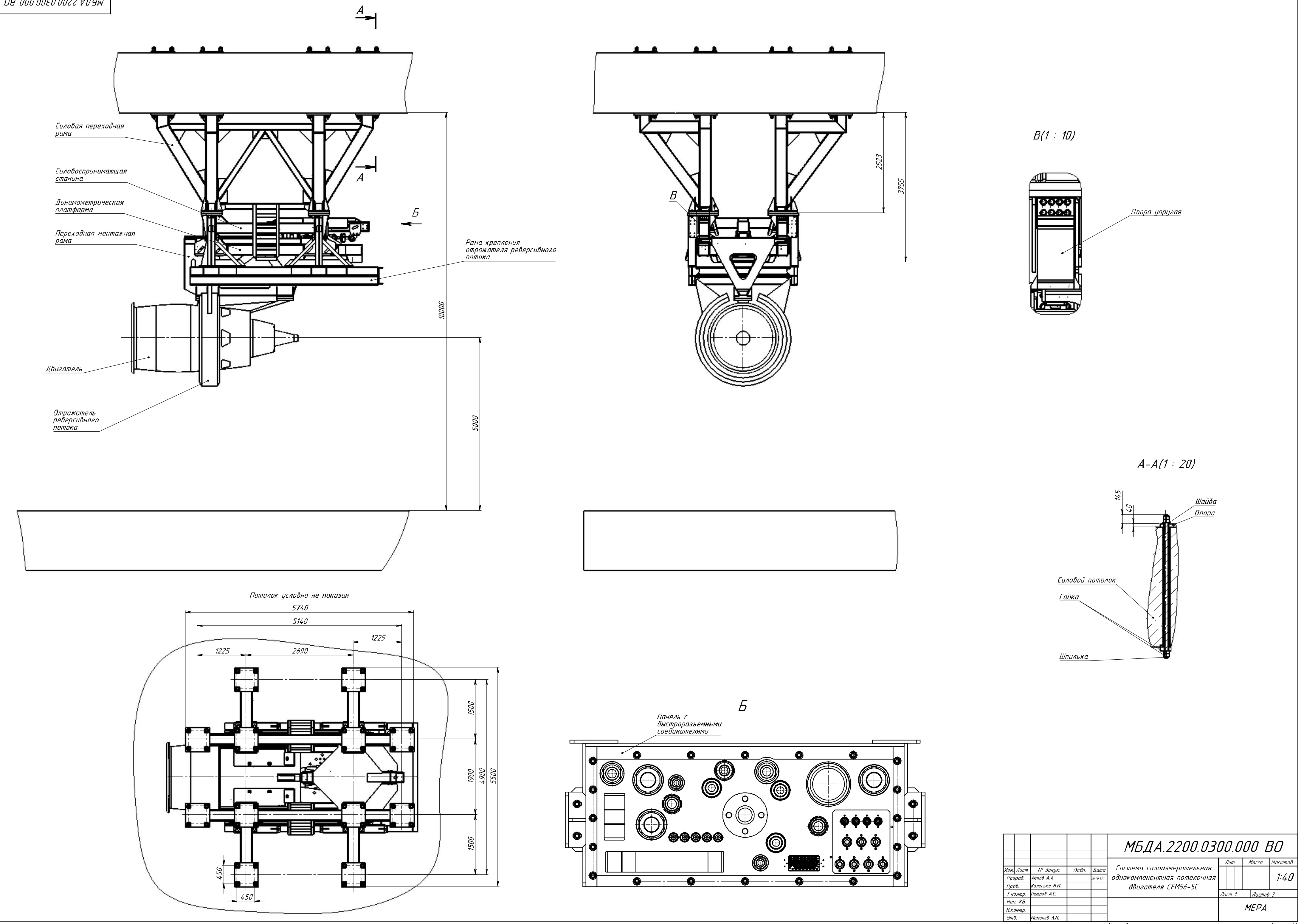
Технологическая компоновка. Поперечный разрез здания
Концептуальный проект. Раздел 3.4
Test bench process arrangement. Cross-section drawing
Conceptual design. Part 3.4

Испытательная станция авиационного двигателя. Технологическая компоновка.
Test bench. Process arrangement

Продольный разрез здания.
Longitudinal section drawing .



Технологическая компоновка. Продольный разрез здания
Концептуальный проект Раздел 3.4.
Process arrangement. Longitudinal section drawing
Conceptual design. Part 3.4.



МБДА.2200.0300.000 ВО

Система силоизмерительная однокомпонентная потолочная двигателя CFM56-5C

Изм.	Лист	№ докум.	Подп.	Дата
Разраб.		Аепов А.А.		26.10.17
Пров.		Колесько М.М.		
Т.контр.		Петров А.С.		
Нач. КБ				
Н.контр.				
Утв.		Мамонов А.М.		

Лит.	Масса	Масштаб
		1:40

Лист 1 | Листов 3

МЕРА

Копировал | Формат А1

Перв. примен. | Справ. № | Подпись и дата | Инв. № дубл. | Взам. инв. № | Подпись и дата | Инв. № подл.

APPENDIX "B"

FIRE HAZARD STANDARDS AND TABLES

B.1 DETERMINATION OF CATEGORIES OF ROOMS, BUILDINGS AND EXTERNAL INSTALLATIONS ON EXPLOSION AND FIRE HAZARD

Information about the set of rules:

1. Developed by "ФГБУ ВНИИПО МЧС[1]" of Russia.

2. Introduced by the Technical Committee for Standardization of TC 274 "Fire Safety".

3. Approved and put into effect by the Order of the Ministry of Emergency Measures of Russia from March 25, 2009 N 182.

4. Registered by the Federal Agency for Technical Regulation and Metrology.

References:

The present set of rules uses references to the following standard:

The system of labor safety standards. Fire and explosion hazard of substances and materials. Nomenclature of indicators and methods for their determination.

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1 Federal State Establishment All-Russian Research Institute for Fire Protection of Ministry of Russian Federation for Civil Defense, Emergencies and Elimination of Consequences of Natural Disasters

Категории помещений	Характеристика веществ и материалов, находящихся (обращающихся) в помещении
А повышенная взрывопожаро-опасность **increased fire and explosion hazard**	Горючие газы, легковоспламеняющиеся жидкости с температурой вспышки не более 28 °С в таком количестве, что могут образовывать взрывоопасные парогазовоздушные смеси, при воспламенении которых развивается расчетное избыточное давление взрыва в помещении, превышающее 5 кПа, и (или) вещества и материалы, способные взрываться и гореть при взаимодействии с водой, кислородом воздуха или друг с другом, в таком количестве, что расчетное избыточное давление взрыва в помещении превышает 5 кПа **Flammable gases, flammable liquids with a flash-point of not more than 28 ° C in such quantity that they can form explosive vapor-gas-air mixtures, upon ignition of which an estimated overpressure of explosion in a room exceeding 5 kPa and / or substances and materials capable of exploding and burn with interaction with water, oxygen or with each other, in such quantity that the calculated excess pressure of the explosion in the room exceeds 5 kPa**
Б взрывопожаро-опасность **explosion hazard**	Горючие пыли или волокна, легковоспламеняющиеся жидкости с температурой вспышки более 28 °С, горючие жидкости в таком количестве, что могут образовывать взрывоопасные пылевоздушные или паровоздушные смеси, при воспламенении которых развивается расчетное избыточное давление взрыва в помещении, превышающее 5 кПа **Flammable dusts or fibers, flammable liquids with a flash point of more than 28 ° C, flammable liquids in such quantities that they can form explosive dust-air or vapor-air mixtures, upon ignition of which an estimated overpressure of explosion in a room exceeding 5 kPa**
В1—В4 пожароопасность **fire risk**	Горючие и трудногорючие жидкости, твердые горючие и трудногорючие вещества и материалы (в том числе пыли и волокна), вещества и материалы, способные при взаимодействии с водой, кислородом воздуха или друг с другом только гореть, при условии, что помещения, в которых они находятся (обращаются), не относятся к категории А или Б **Hot and combustible liquids, solid combustible and difficult combustible substances and materials , substances and materials capable of interacting with water, oxygen or with each other only burn, provided that the premises in which they are located (apply), do not belong to category A or B**
Г умеренная пожароопасность **moderate fire hazard**	Негорючие вещества и материалы в горячем, раскаленном или расплавленном состоянии, процесс обработки которых сопровождается выделением лучистого тепла, искр и пламени, и (или) горючие газы, жидкости и твердые вещества, которые сжигаются или утилизируются в качестве топлива **Non-combustible substances and materials in hot, hot or molten state, the processing of which is accompanied by the release of radiant heat, sparks and flames, and (or) combustible gases, liquids and solids that are burned or utilized as fuel**
Д пониженная пожароопасность **reduced fire hazard**	Негорючие вещества и материалы в холодном состоянии **Non-flammable substances and materials in a cold state**

B.2 RULES FOR THE INSTALLATION OF ELECTRICAL INSTALLATIONS

EIR (Electrical Instalation Requirments) is designed taking into account the mandatory implementation in the operating conditions of preventive tests, repairs of electrical installations, and their electrical equipment.

It is developed taking into account the requirements of state standards, building rules, recommendations of scientific and technical councils for review of draft chapters. The drafts of the chapters were reviewed by the working groups of the Coordinating Council for the revision of the EIR prepared by ОАО "ВНИИЭ[2]" and agreed in accordance with the established procedure with Gosstroy of Russia, Gosgortekhnadzor of Russia, РАО «ЕЭС России» (ОАО ВНИИЭ) and submitted for approval by Gosenergonadzor of the Ministry of Energy of Russia.

These requirements approved by the Ministry of Energy of the Russian Federation, Order No. 204 of July 8, 2002.

2 All-Union Scientific Research Institute of Electric Power Engineering

Зона по ПУЭ **Zone according to EIR**	Характеристика веществ и материалов, находящихся (обращающихся) в помещении **Characteristics of substances and materials that are (circulating) indoors**
П-I Пожароопасная **fire-hazardous**	горючие жидкости с температурой вспышки > 61 °C **flammable liquids with a flash-point> 61 ° C**
П-II Пожароопасная **fire-hazardous**	горючие пыль волокна с нижним концентрационным пределом воспламенения более **65 г/м^3** **Flammable dust fibers with a lower concentration limit of ignition greater than 65 g / m3**
П-IIa Пожароопасная **fire-hazardous**	твердые горючие вещества **solid combustibles**
П-III Пожароопасная **fire-hazardous**	горючие жидкости с температурой вспышки **> 61 °C** твердые горючие вещества (вне помещений) **flammable liquids with a flash-point> 61 ° C** **solid flammable substances (outdoor)**
В-I Пожароопасная **explosive**	горючие газы или пары **ЛВЖ** (Легковоспламеняющаяся жидкость) могут образовать с воздухом взрывоопасные смеси при нормальных режимах работы **Flammable gases or VHF (Highly flammable liquid) vapors may form explosive mixtures with air under normal operating conditions**
В-Ia Пожароопасная **explosive**	горючие газы или пары **ЛВЖ** могут образовать с воздухом взрывоопасные смеси при аварии **Flammable gases or VHF vapors may form explosive mixtures with air in an accident**
В-Iб Пожароопасная **explosive**	горючие газы или пары **ЛВЖ** могут образовать с воздухом взрывоопасные смеси при аварии аммиачные установки**,** электролиз воды, зарядные станции аккумуляторных батарей и т.п. **Flammable gases or VHF vapors may form explosive mixtures with air in an accident** **ammonia plants, electrolysis of water, charging stations of storage batteries, etc.**